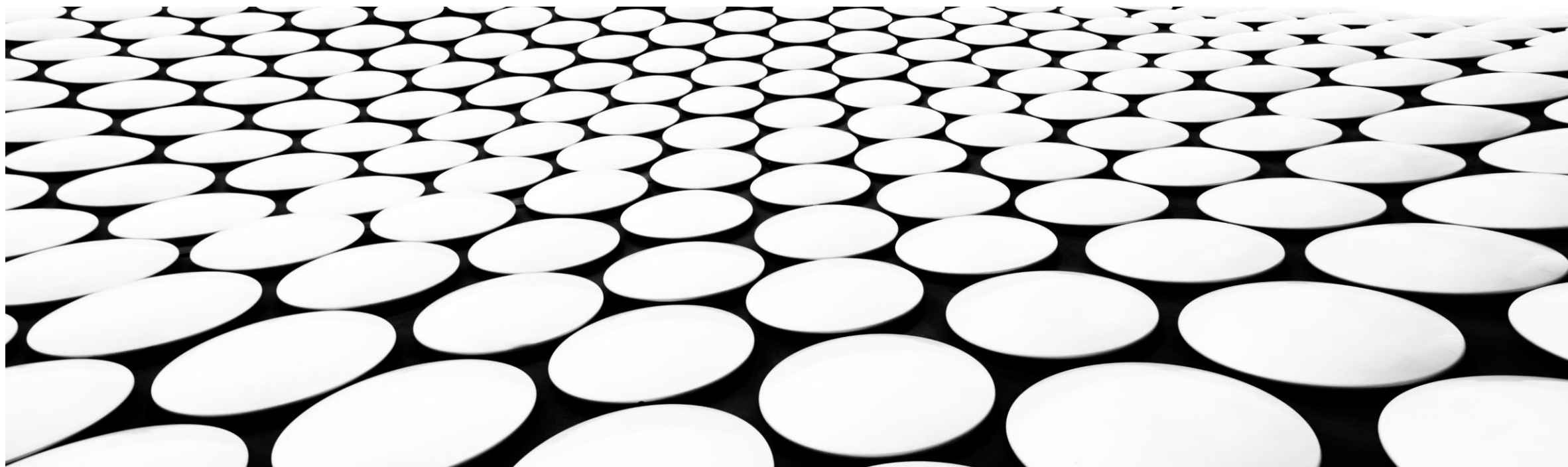

WHEN MACHINES LEARNED TO FIND MEANING ACROSS LANGUAGES

DR. PINAKI BHASKAR

MACHINE TRANSLATION IN BENGALI: HISTORY, METRICS, AND THE AI FRONTIER – DAY 3





PHASE II: WHEN MACHINES LEARNED TO FIND MEANING ACROSS LANGUAGES

- Workshop: “Machine Translation in Bengali: History, Metrics, and the AI Frontier”
- Phase II: Translation-adjacent systems for Bengali
- Focus today: cross-lingual access, summarisation, and question answering
- Resource person: **Dr. Pinaki Bhaskar**, Samsung R&D Institute India, Bangalore



THE FIVE-DAY JOURNEY: WHERE WE ARE

- Days 1–2: Foundational infrastructure – script, OCR, corpora (Prof. B. B. Chaudhuri)
- **Days 3–4: Meaning across languages & neural MT (my sessions)**
- Day 5: LLMs, agents, and the contemporary frontier (Pritam Dey)
- Today: the “bridge” period between infrastructure and neural translation



MY PATH INTO BENGALI LANGUAGE TECHNOLOGY

- PhD, Jadavpur University: Multi-document Summarisation and QA for English & Bengali
- Postdoc, CNR Pisa: Biomedical Question Answering with IR, IE, and Knowledge Graphs
- Now at Samsung: GenAI, VLMs, and Agentic AI for mobile experiences
- I work on systems that **find and organise meaning**, not just translate sentences



WHAT IS NATURAL LANGUAGE PROCESSING (NLP)?

- The process of building computational models for understanding natural language.
- INPUT: natural language text.
- OUTPUT: representation of the meaning of the text.
- Sometimes called natural language understanding (NLU) or computational linguistics.

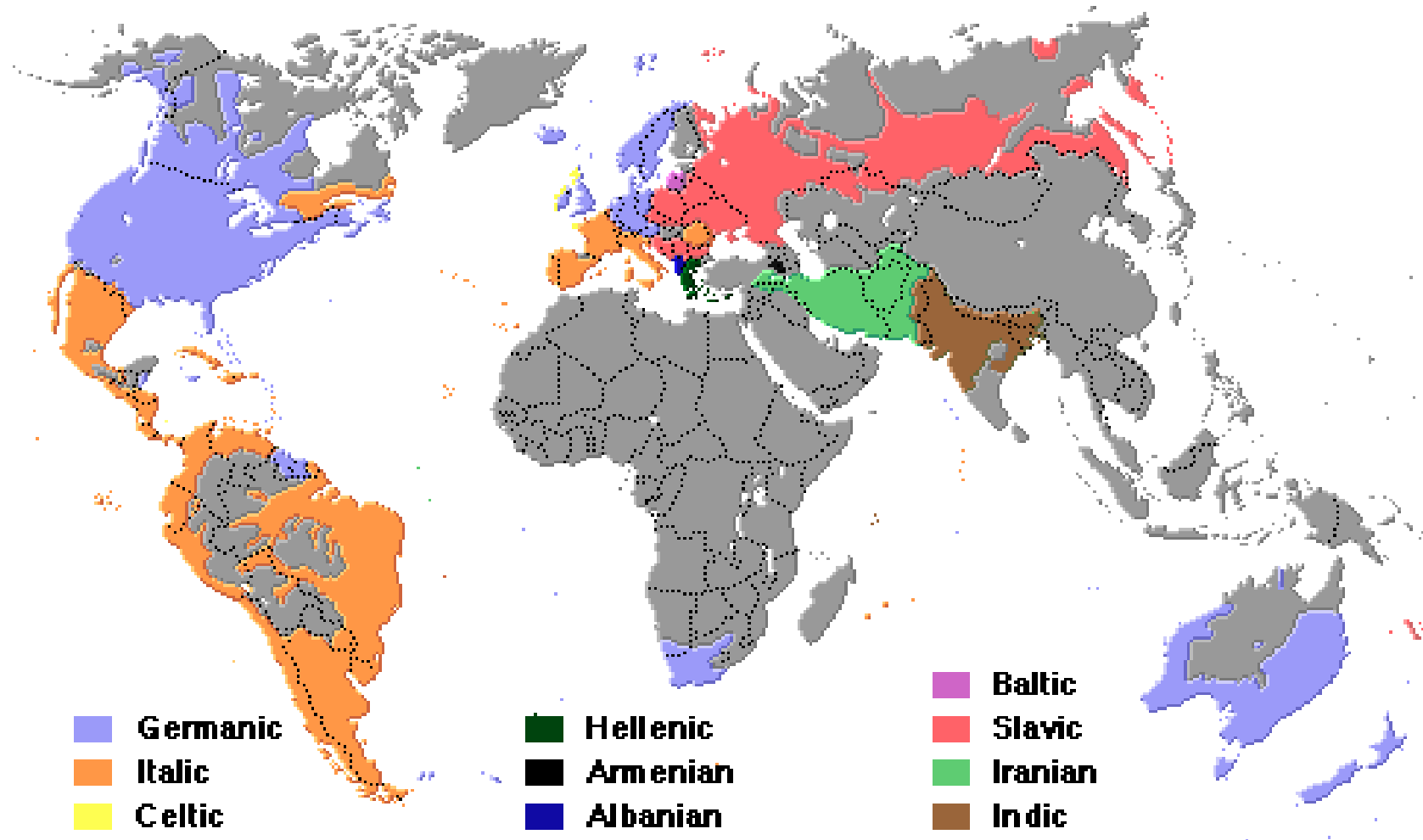
NATURAL LANGUAGE PROCESSING

- Computational models of aspects of human language processing:
 - Understanding a textbook
 - Writing a letter
 - Carrying on a conversation
 - Translating a document
 - Searching for information

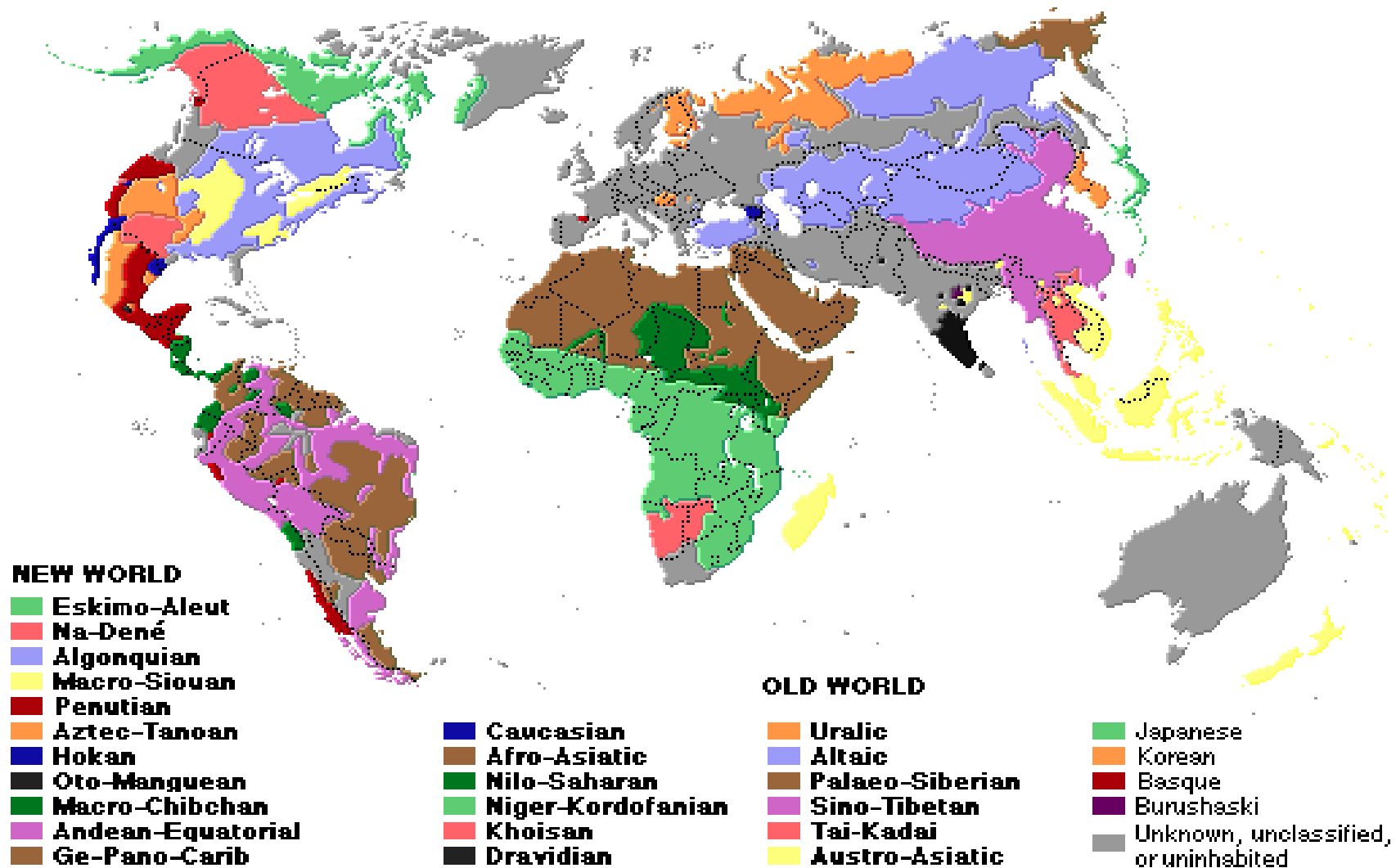
HUMAN LANGUAGES

- 6809 known spoken human languages (debated)
- 4% of languages spoken by 97% of world's population
- 96% of languages spoken by 3% of world
- 10% on Internet
- 48 different first languages in Israel
- Mandarin Chinese has 845,000,000 speakers
- 543 extinct languages (at least)
- 417 on verge of extinction (2 speak Ter Sami)
- Half on way to extinction (spoken by <10,000)

INDO-EUROPEAN



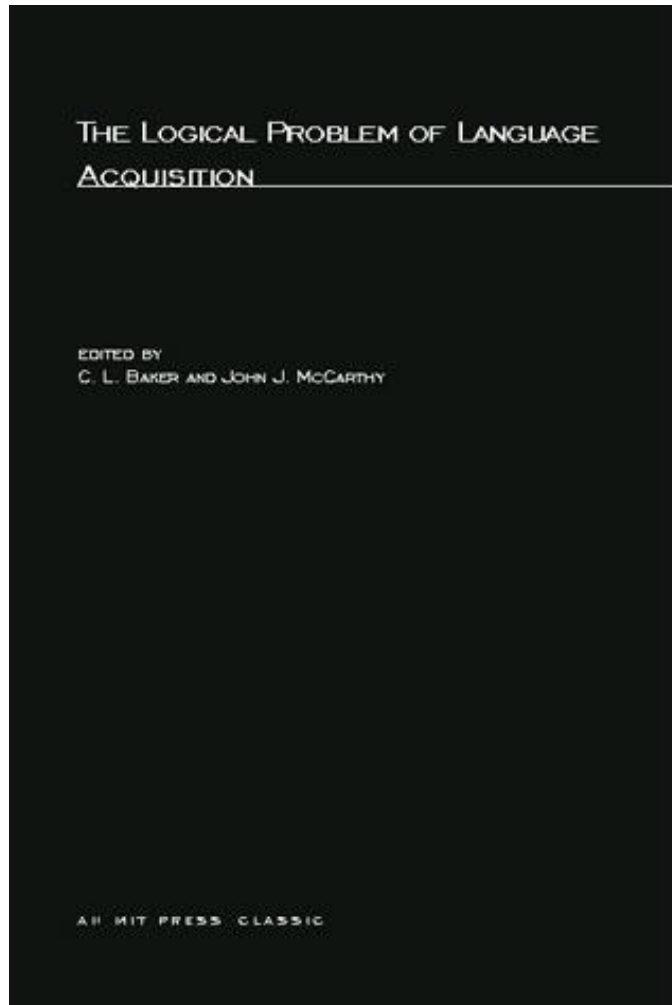
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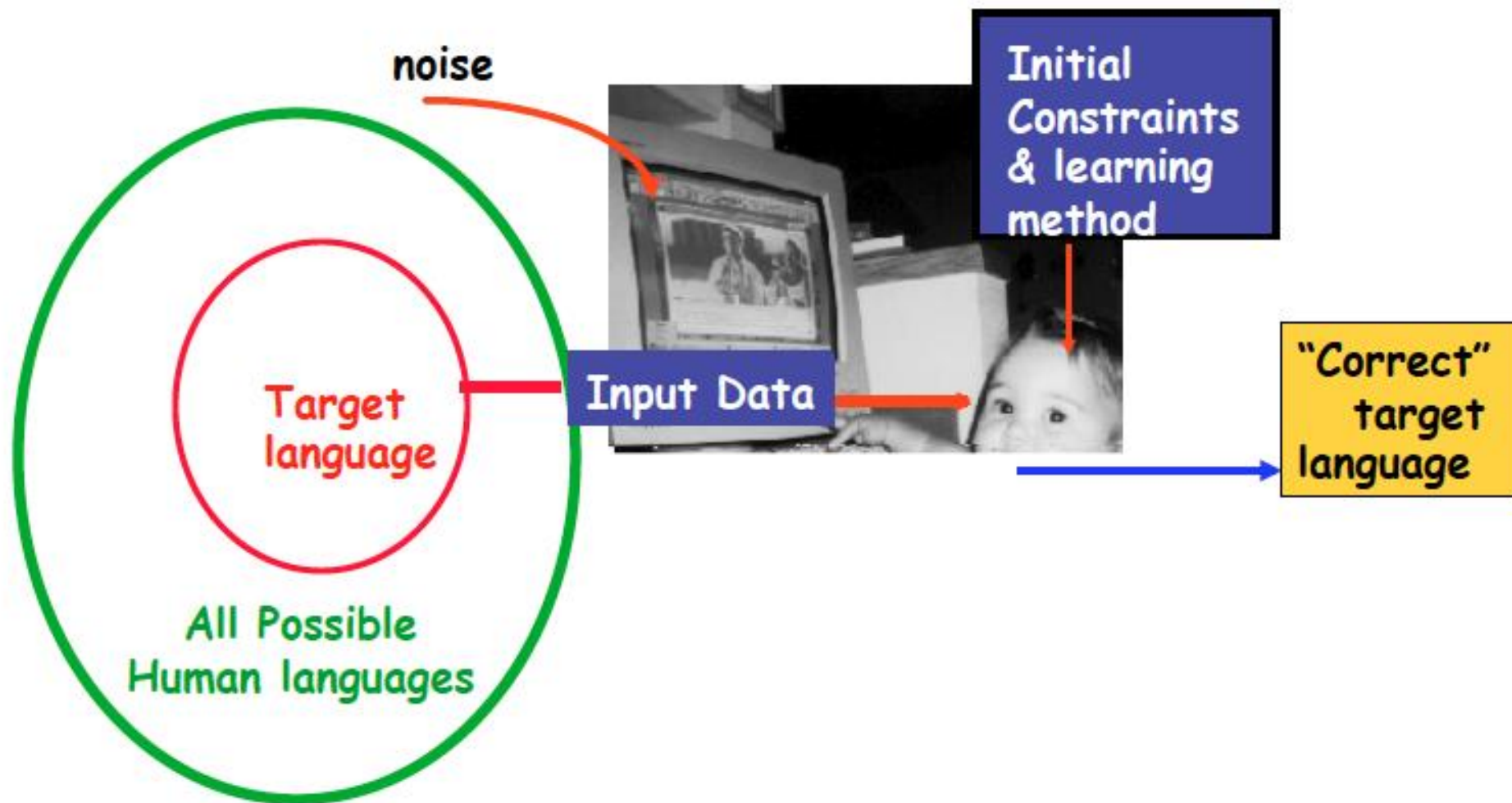
HUMAN LANGUAGES

- You know ~50,000 words of primary language, each with several meanings
- Six year old knows ~13000 words
- First 16 years we learn 1 word every 90 minutes of waking time
- Mental grammar generates sentences
 - virtually every sentence is novel
- 3 year olds already have 90% of grammar

THE LOGICAL PROBLEM OF LANGUAGE ACQUISITION



- “problem of discovering the character of the mental capacities that make it possible for human beings to attain knowledge of their language on the basis of fragmentary and haphazard early linguistic experience”



HUMAN SPOKEN LANGUAGE

- Most complicated mechanical motion of the human body
 - ▣ Movements must be accurate to within mm
 - ▣ synchronized within 100ths of a second
- We can understand up to 50 phonemes/sec (normal speech 10-15ph/sec)
 - ▣ but if sound is repeated 20 times /sec we hear continuous buzz!
- All aspects of language processing are involved and manage to keep apace

THIS MODEL SHOWS WHAT A MAN'S BODY WOULD LOOK LIKE IF EACH PART GREW IN PROPORTION TO THE AREA OF THE CORTEX OF THE BRAIN CONCERNED WITH ITS MOVEMENT.



NATURAL LANGUAGE PROCESSING

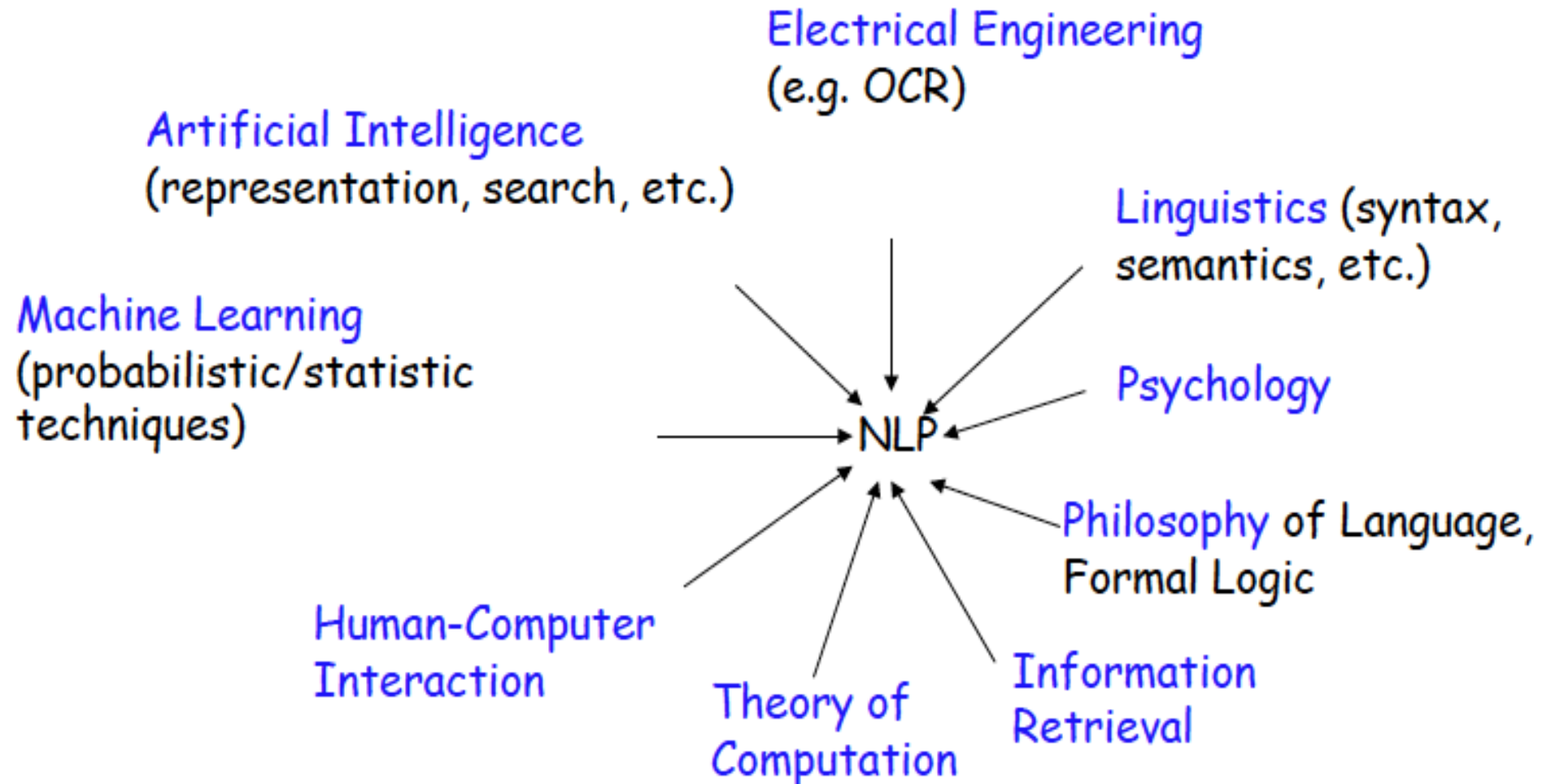
- ☐ Speech Recognition
- ☐ Natural Language Understanding (NLU)
- ☐ Computational Linguistics (CL)
- ☐ Psycholinguistics
- ☐ Information Extraction (IE)
- ☐ Information Retrieval (IR)
- ☐ Natural Language Generation (NLG)
- ☐ Speech Synthesis
- ☐ Language Evolution etc.

APPLIED NLP

- Machine Translation (MT)
- Spelling/grammar correction
- Information Retrieval / Search Engines :
Retrieval, Categorization, Routing, Filtering, Summarization
- Data mining
- Document classification
- Question Answering (QA), conversational agents
- Plagiarism detection

and many more....

RELATED DISCIPLINES



LEVELS OF ANALYSIS AND KNOWLEDGE USED IN NLP

- ❑ **Morphology:** how words are constructed; prefixes & suffixes
- ❑ **Syntax:** structural relationships between words
- ❑ **Semantics:** meanings of words, phrases, and expressions
- ❑ **Discourse:** relationships across different sentences or thoughts; contextual effects
- ❑ **Pragmatic:** the purpose of a statement; how we use language to communicate
- ❑ **World Knowledge:** facts about the world at large; common sense

MORPHOLOGICAL ANALYSIS

- Inflectional
 - duck + s = [N duck] + [plural s]
 - duck + s = [V duck] + [3rd person s]
- • Derivational
 - kind, kindness
- • Spelling changes
 - drop, dropping
 - hide, hiding

MORPHOLOGY HUMOR



I WATCHED THE TERRAPIN.



Stu Weiss

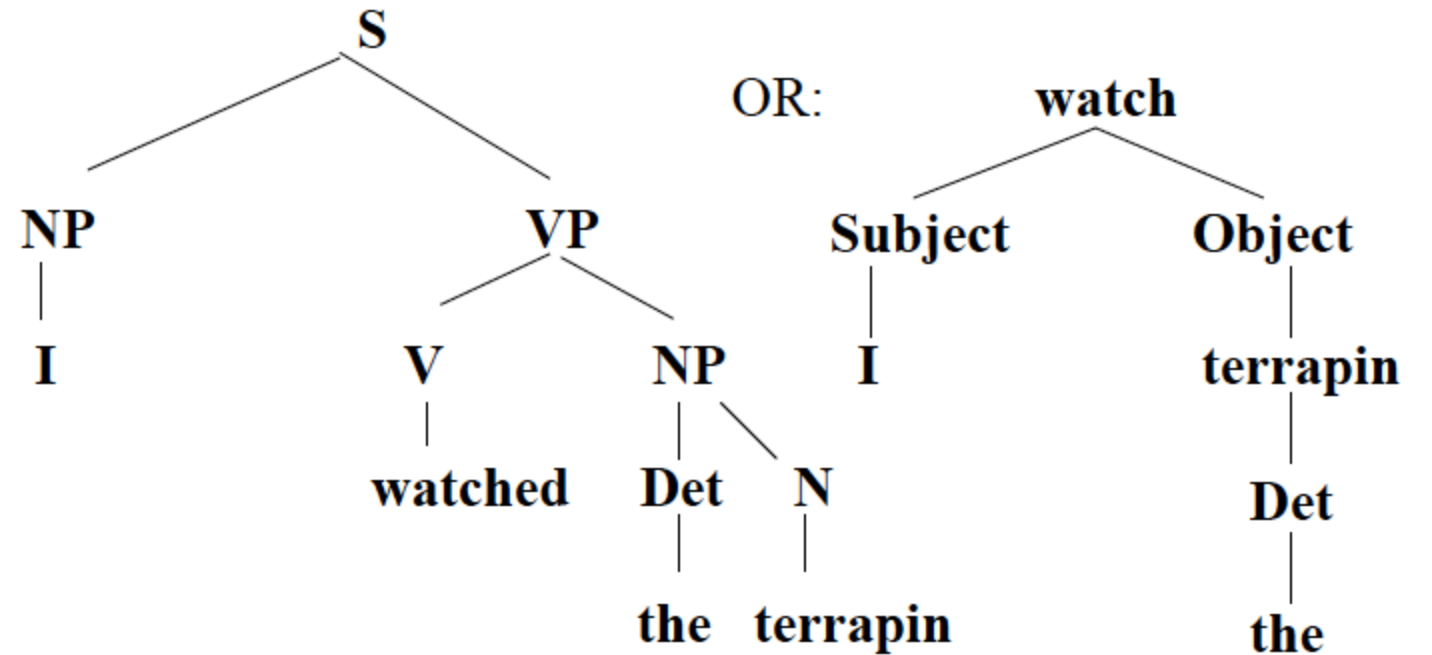
8-May-26

SYNTAX

- SVO common: English, French, Danish, Chinese, Swahili (Tanzania)
- SOV common: German, Turkish, Japanese, Persian, Korean
- VSO rare: Gaelic (Ireland), Arabic, Welsh (UK)
- VOS rare: Mopán Maya (Belize), Bushi (Madagascar), Fijian (Fiji)
- OSV extremely rare: Xavante (Brazil)
- OVS extremely rare: Panare (Venezuela), Macushi (Guyana), [Klingon]

SYNTACTIC ANALYSIS

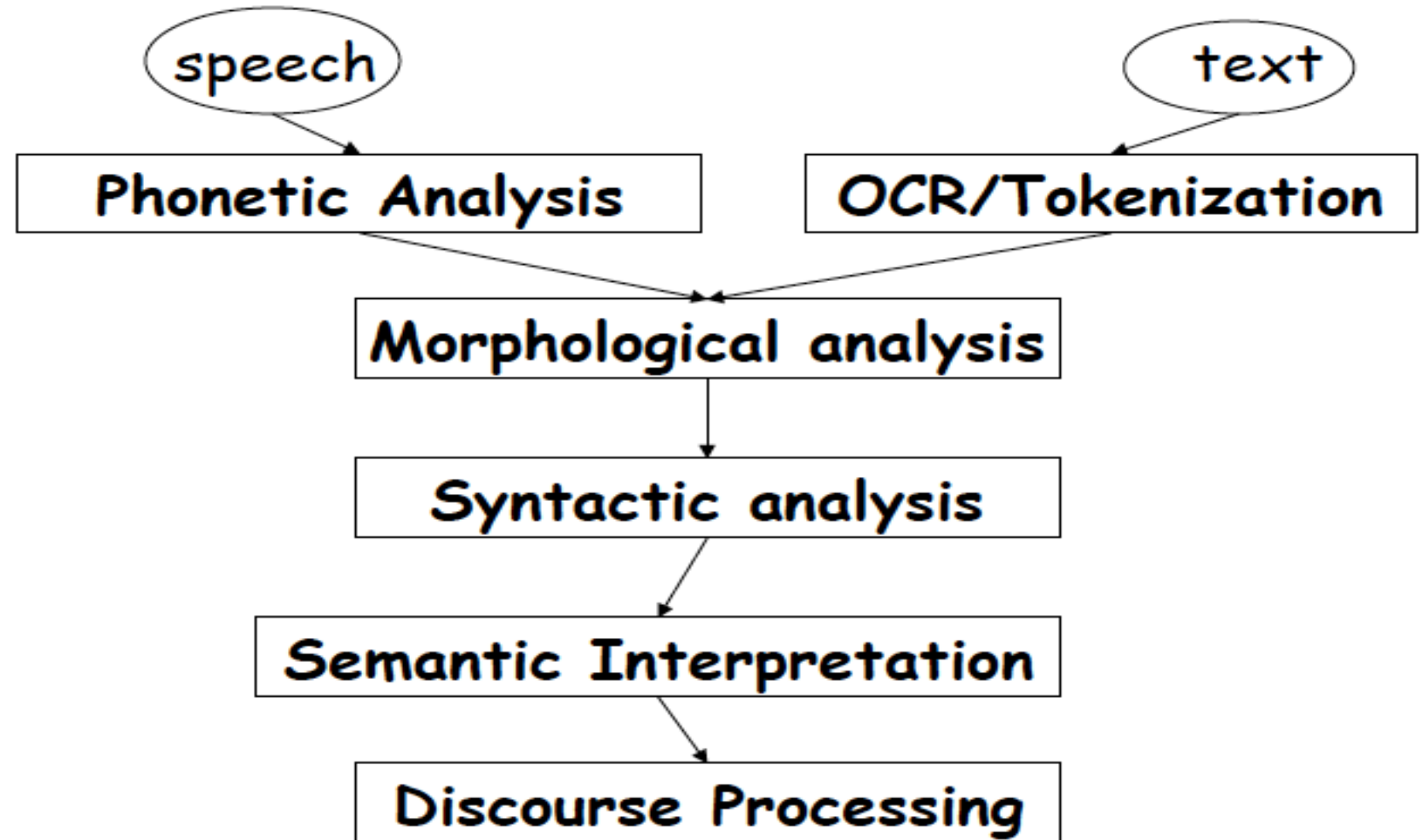
- Associate constituent structure with string
- Prepare for semantic interpretation



SEMANTIC INTERPRETATION

- A way of representing meaning
- Abstracts away from syntactic structure
- Example:
 - First-Order Logic: `watch(I, terrapin)`
 - Can be: “I watched the terrapin” or “The terrapin was watched by me”
- Real language is complex:
 - What did I watch?

NLP PIPELINE





NL UNDERSTANDING

Query:

“Find me all articles concerning car accidents involving more than two cars in Malta during the first half of last year.”

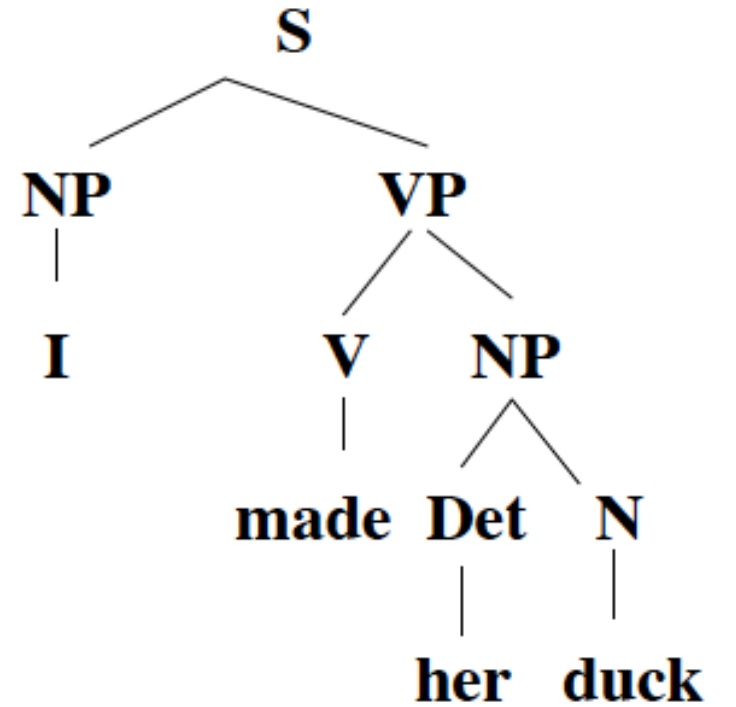
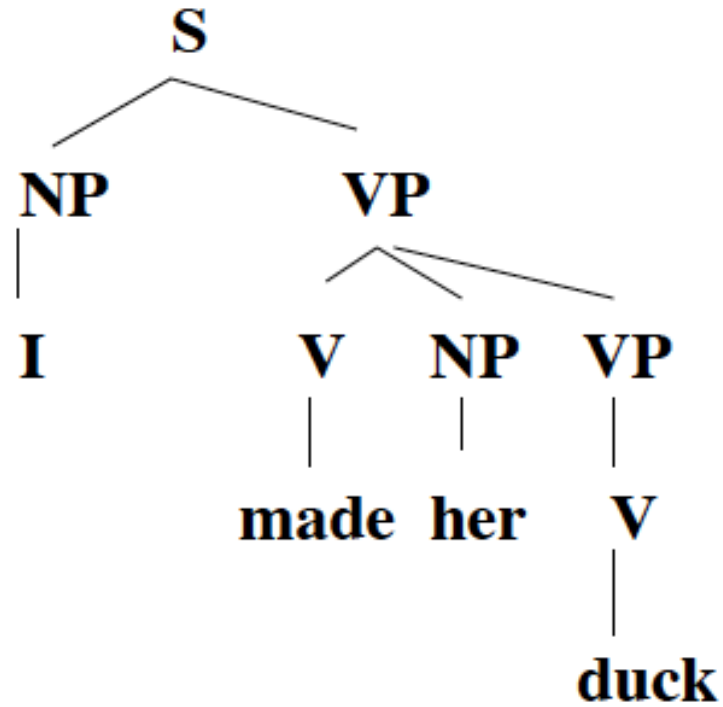
System must extract enough information to determine whether the article meets the criterion defined by the query

SYNTAX: STRUCTURAL AMBIGUITY

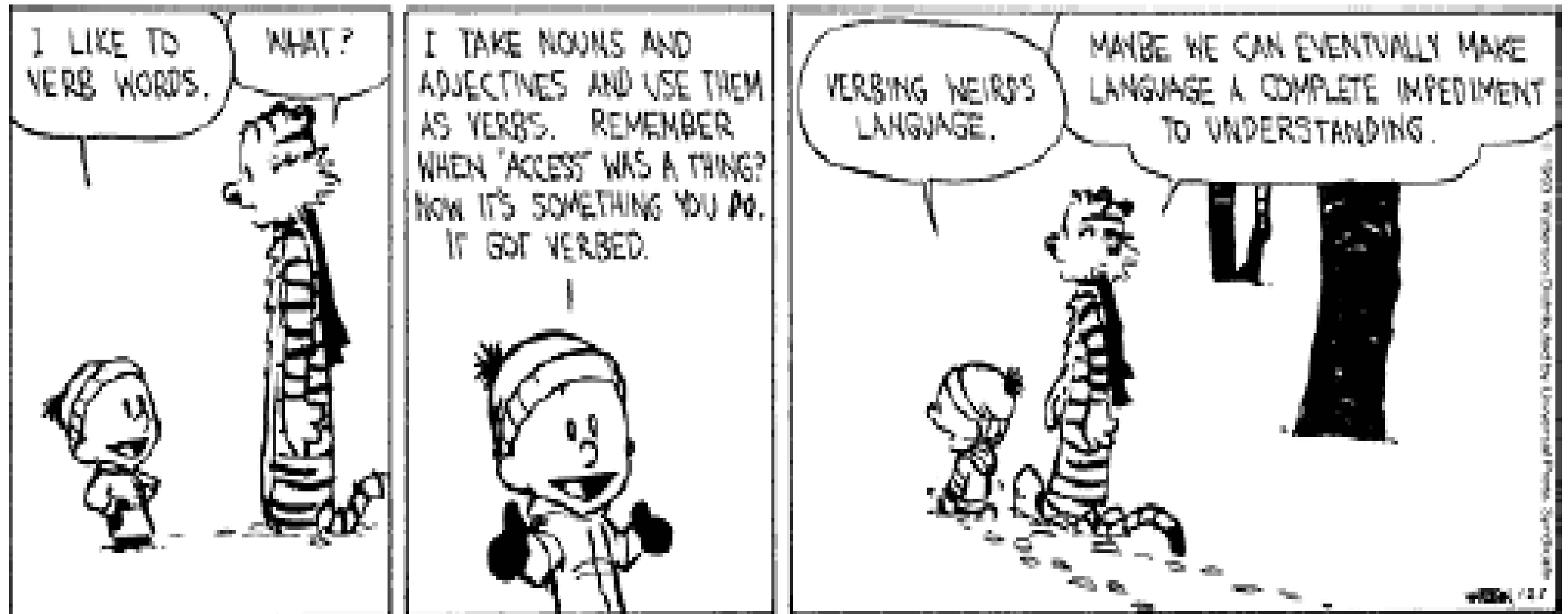
- ☐ I made her duck
- ☐ I made duckling for her
- ☐ I made the duckling belonging to her
- ☐ I created the duck she owns
- ☐ I forced her to lower her head
- ☐ By magic, I changed her into a duck

SYNTACTIC DISAMBIGUATION

- Structural ambiguity:



LANGUAGE UNDERSTANDING HUMOR



SEMANTICS: LEXICAL AMBIGUITY

- I walked to the bank ...
 of the river.
 to get money.
- The bug in the room ...
 was planted by spies.
 flew out the window.
- I work for John Hancock ...
 and he is a good boss.
 which is a good company.

WORD SENSE HUMOR



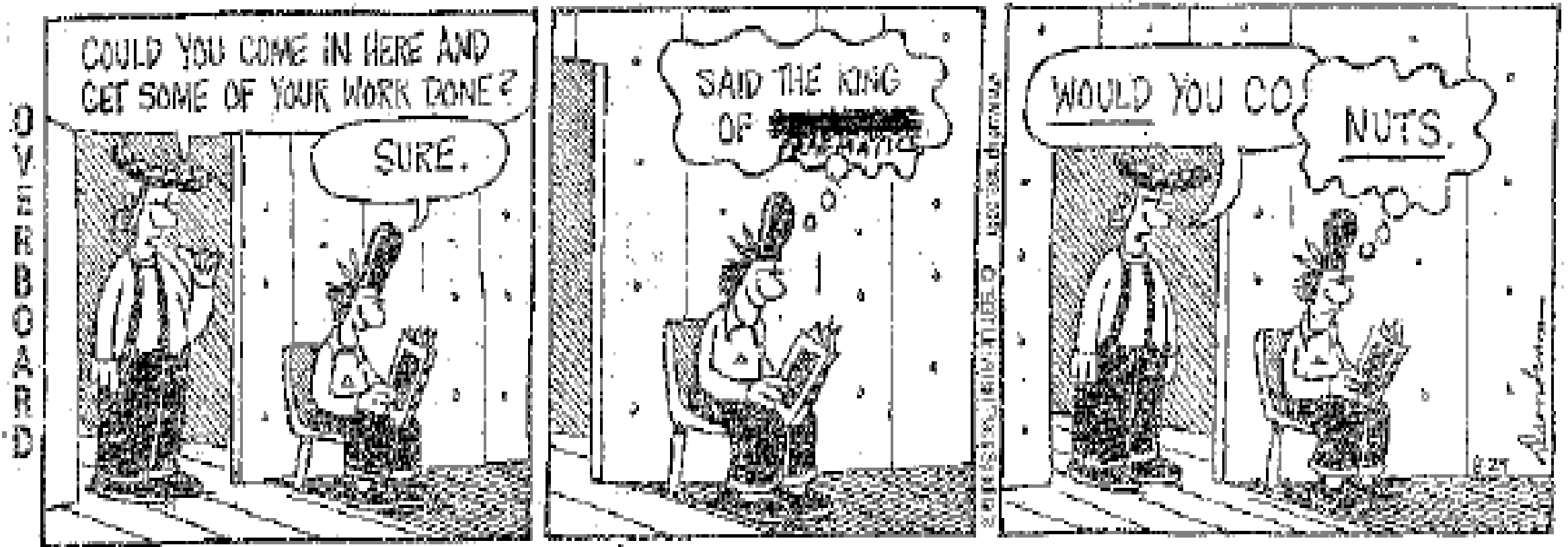
DISCOURSE: ANAPHORA & COREFERENCE

- President John F. Kennedy was assassinated.
- The president was shot yesterday.
- Relatives said that John was a good father.
- JFK was the youngest president in history.
- His family will bury him tomorrow.
- Friends of the Massachusetts native will hold a candlelight service in Mr. Kennedy's home town.

PRAGMATICS

- Rules of Conversation
 - Can you tell me what time it is?
 - Could I please have the salt?
- Speech Acts
 - I bet you \$50 that the Jazz will win.
 - Will you marry me?

PRAGMATICS HUMOR





WORLD KNOWLEDGE

- John went to the diner. He ordered a steak. He left a tip and went home.
- John wanted to commit suicide. He got a rope.



BREAK

FOR 10 MINS



TRANSLATION: FROM “RIGHT WORD” TO “RELEVANT MEANING”

- Classical view: translation = finding the “right” equivalent words
- Example: “freedom” → “স্বাধীনতা” or “মুক্তি” – context decides
- Modern systems often first find **relevant documents and passages**
- In practice, translators already do this: search, skim, then craft language

HOW A SEARCH ENGINE READS BENGALI TEXT

- Computers store documents as indexed word statistics, not as stories
- Query is treated as a short text to match against this index
- Relevance is computed from shared words, frequency, and patterns
- This is the backbone of information retrieval (IR) systems
- **Example snippet (Bengali news headline):**
 - “কলকাতায় গঙ্গা দূষণ নিয়ে নতুন রিপোর্ট প্রকাশ”

ACCESSING OTHER-LANGUAGE TEXTS: CLIR AND CLIA

- IR: query and documents in **one** language
- CLIR: query in one language, documents in another (e.g., English → Bengali)
- CLIA: adds snippets, summaries, and QA on top of cross-lingual retrieval
- For Bengali, CLIA means: English question, Bengali sources, bilingual answers
- **Example query:**
 - English: “river pollution near Kolkata in 2020”
 - Bengali documents returned with headlines like above



WHY RETRIEVAL CAME BEFORE FLUENT TRANSLATION

- In low-resource settings, it was often easier to retrieve relevant documents across languages than to generate high-quality target sentences.
- This led to CLIR and CLIA systems that helped users *find* information before full MT matured.
- Conceptually, this shifts the goal from “exact translation” to “cross-lingual access.”



THE INDIAN CLIA INITIATIVE

- National project: Cross-Lingual Information Access for Indian languages
- Supports querying in one language and accessing many Indian languages
- Bengali included along with Hindi, Tamil, Telugu, etc.
- Built by a consortium of institutes; seeded many later research efforts

ENGLISH–BENGALI CLIA: A SIMPLIFIED PIPELINE

- Step 1: User types query in English
- Step 2: System expands / translates query into Bengali representations
- Step 3: Bengali document retrieval over news, reports, archives
- Step 4: System shows snippets, summaries, or answer candidates
- **Concrete example:**
 - Query: “How is air quality in Kolkata this winter?”
 - Bengali snippet: “এই শীতে কলকাতার বায়ুর মান বিপজ্জনক মাত্রার কাছাকাছি পৌঁছেছে...”

ACTIVITY: “WHERE WOULD YOU SEARCH?”

- In pairs: Person A writes an **English information need**
- Person B lists Bengali keywords, sources, or websites they would search
- Then actually search: what do Google / news portals show?
- Discuss: what was missing, distorted, or surprisingly relevant?
- **Suggested Bengali information need to seed groups:**
 - “২০১৮ থেকে ২০২৪ পর্যন্ত সুন্দরবনের ঘূর্ণিঝড়ের প্রভাব”



WHY SUMMARIZATION MATTERS FOR TRANSLATION

- Real-world topics span dozens of articles, not one canonical text
- Translators often first ask: “What’s going on overall?”
- Single-document vs multi-document summarization
- Summaries guide what deserves careful translation and what can be skimmed

MANY REPORTS, ONE STORY (EXAMPLE)

- Suppose we have three Bengali news pieces on the same protest
- Headlines (fabricated but realistic):
 - “চাকরি প্রার্থীদের প্রদর্শনে কলকাতায় যানজট”
 - “বেতন কাঠামো নিয়ে শিক্ষকদের অবস্থান বিক্ষোভ”
 - “শহরের বিভিন্ন জায়গায় শিক্ষকদের মিছিল, সাধারণ মানুষের ভোগান্তি”
- A multi-document summary could be:
 - “বেতন কাঠামো ও চাকরির দাবিতে শিক্ষকদের একাধিক সংগঠন কলকাতার বিভিন্ন জায়গায় মিছিল ও অবস্থান বিক্ষোভ করায় শহরে ব্যাপক যানজট ও ভোগান্তি তৈরি হয়েছে।”
- **Talking point:** what nuance and local colour were lost in the summary?

QUESTION ANSWERING: ASKING MACHINES DIRECT QUESTIONS

- QA systems take a question and return answer spans from documents
- They reduce the burden of reading entire articles
- For Bengali, QA can operate over Bengali or mixed Bengali–English corpora
- QA is a form of “targeted translation of information needs”
- **Example:**
 - Question: “এই বিক্ষোভের প্রধান দাবি কী ছিল?”
 - Answer span from article: “প্রাথমিক শিক্ষকদের বেতন কাঠামো সংশোধনের দাবি ছিল আন্দোলনের মূল কেন্দ্রবিন্দু।”



MEASURING SYSTEM QUALITY: WHY METRICS?

- Research and deployment need **comparisons** across systems
- Retrieval: precision, recall, F-score, MAP, nDCG
- Translation: BLEU, chrF, COMET and others
- Metrics give quick signals but do not replace human judgement

BLEU AND FRIENDS WITHOUT FORMULAS

- BLEU: counts overlapping word-sequences (n-grams) with reference translations
- chrF: similar, but uses characters – kinder to inflected forms
- COMET: neural model trained to predict human quality scores
- All assume that reference translations are trustworthy ground truth

Toy example (Bengali → English):

- Source (Bengali):
 - “সে ধীরে ধীরে ঘরের দিকে হাঁটল।”
 - Candidate 1: “He walked slowly towards the house.”
Candidate 2: “He strolled back home.”
- BLEU will likely favour Candidate 1 because of n-gram overlap, even if Candidate 2 is stylistically better.

WHY STANDARD METRICS STRUGGLE WITH BENGALI

- Bengali has rich morphology and flexible word order
- Multiple registers: চলিত, সাধু, regional varieties, code-mixing
- Example paraphrases with same meaning but different forms:
 - “সে খুব ক্লান্ত ছিল, কিন্তু কাজটা শেষ করল।”
 - “ভীষণ ক্লান্ত থাকা সত্ত্বেও সে কাজটা শেষ করে ফেলল।”
- Metrics often penalise such paraphrases even when humans see them as equivalent



DAY 3: TRANSLATION AS ACCESS, NOT JUST OUTPUT

- CLIA, summarisation, and QA organise meaning across languages
- Automatic metrics give useful but incomplete signals for Bengali
- Many “translation problems” begin as **information access** problems
- Tomorrow: how neural MT and LLMs actually generate Bengali translations



BREAK

UNTIL 2:15 PM





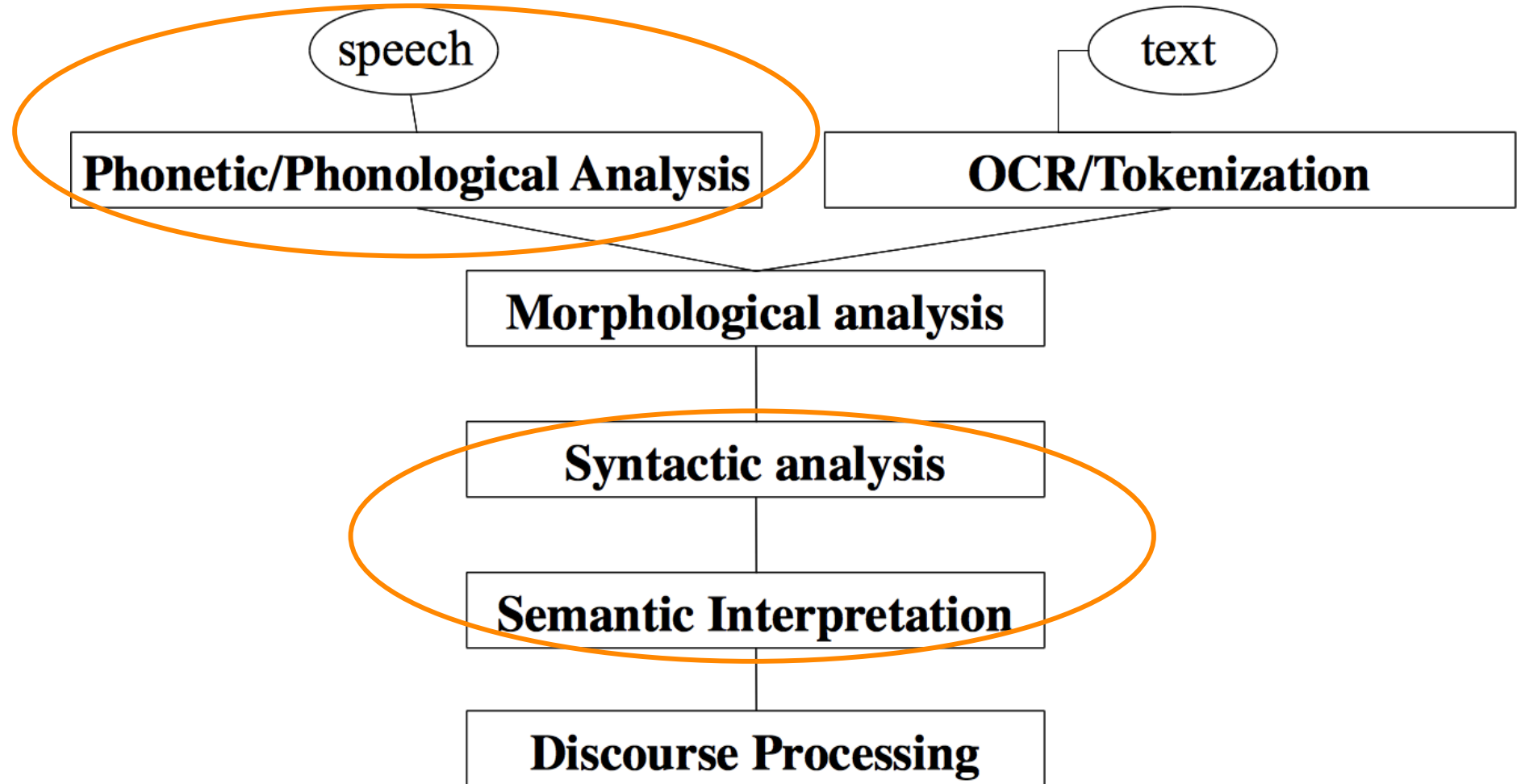
WHAT IS ARTIFICIAL INTELLIGENCE?

- According to the father of Artificial Intelligence, John McCarthy, it is “**The science and engineering of making intelligent machines, especially intelligent computer programs**”.
- Artificial Intelligence is a way of making a computer, a computer-controlled robot, or a software **think intelligently, in the similar manner the intelligent humans think**.
- AI is accomplished by studying **how human brain thinks, and how humans learn, decide, and work** while trying to solve a problem, and then using the outcomes of this study as a basis of developing intelligent software and systems.

WHAT IS NATURAL LANGUAGE PROCESSING (NLP)?

- Natural language processing is a field at the intersection of
 - computer science
 - artificial intelligence
 - and linguistics.
- Goal: for computers to process or “understand” natural language in order to perform tasks that are useful, e.g.
 - Question Answering
- Fully **understanding and representing** the **meaning** of language (or even defining it) is an illusive goal.
- Perfect language understanding is AI-complete

NLP LEVELS





(A TINY SAMPLE OF) NLP APPLICATIONS

- Applications range from simple to complex:
- Spell checking, keyword search, finding synonyms
- Extracting information from websites such as
 - product price, dates, location, people or company names
- Classifying, reading level of school texts, positive/negative sentiment of longer documents
- Machine translation
- Spoken dialog systems
- Complex question answering

NLP IN INDUSTRY

- Search (written and spoken)
- Online advertisement
- Automated/assisted translation
- Sentiment analysis for marketing or finance/trading
- Speech recognition
- Automating customer support





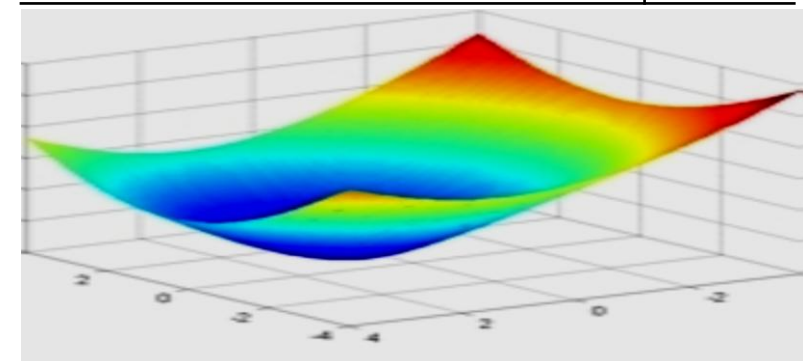
WHY IS NLP HARD?

- Complexity in representing, learning and using linguistic/situational/world/visual knowledge
- Amit hit Rahul and then **he** [fell/ran].
- Ambiguity: “I made her duck”

WHAT'S DEEP LEARNING (DL)?

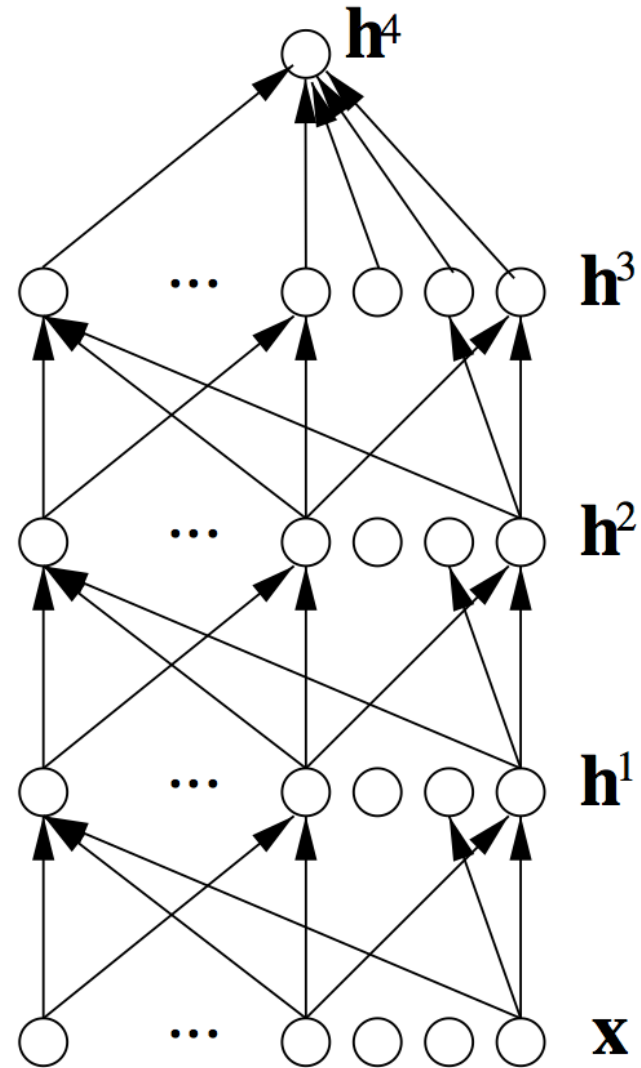
- Deep learning is a subfield of machine learning
- Most machine learning methods work well because of human-designed representations and input features
 - For example: features for finding named entities like locations or organization names (Finkel, 2010):
- Machine learning becomes just optimizing weights to best make a final prediction

Feature	NER
Current Word	✓
Previous Word	✓
Next Word	✓
Current Word Character n-gram	all
Current POS Tag	✓
Surrounding POS Tag Sequence	✓
Current Word Shape	✓
Surrounding Word Shape Sequence	✓
Presence of Word in Left Window	size 4
Presence of Word in Right Window	size 4



WHAT'S DEEP LEARNING (DL)?

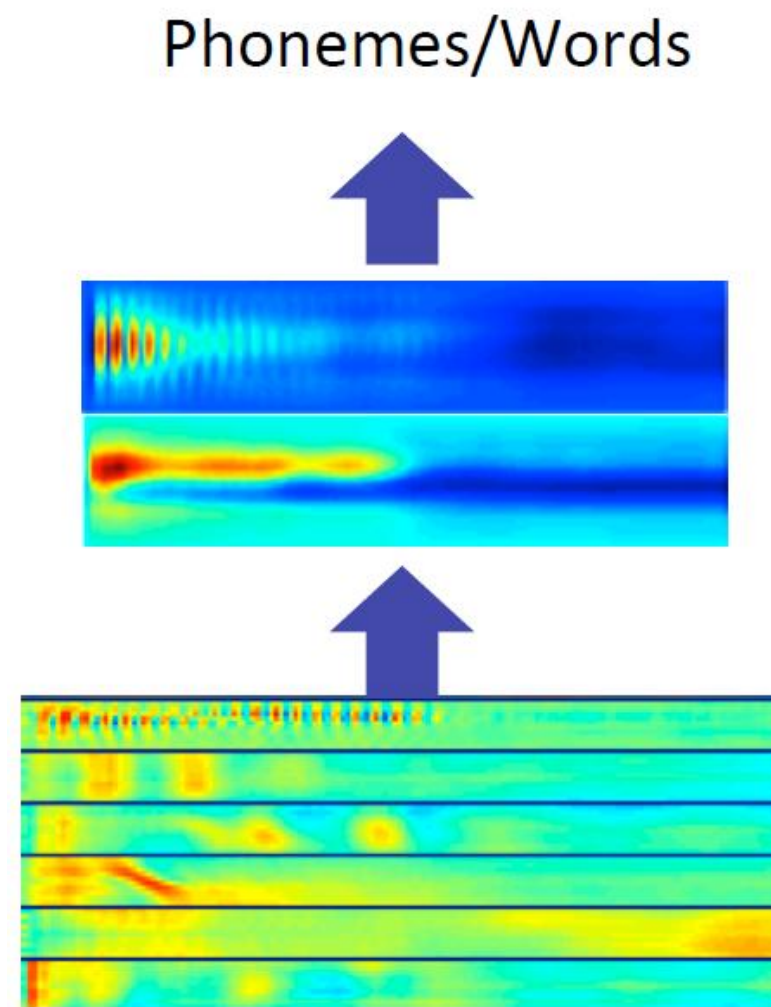
- Representation learning attempts to automatically learn good features or representations
- Deep learning algorithms attempt to learn (multiple levels of) representation and an output
- From “raw” inputs $\mathbf{x}$ (e.g. words)



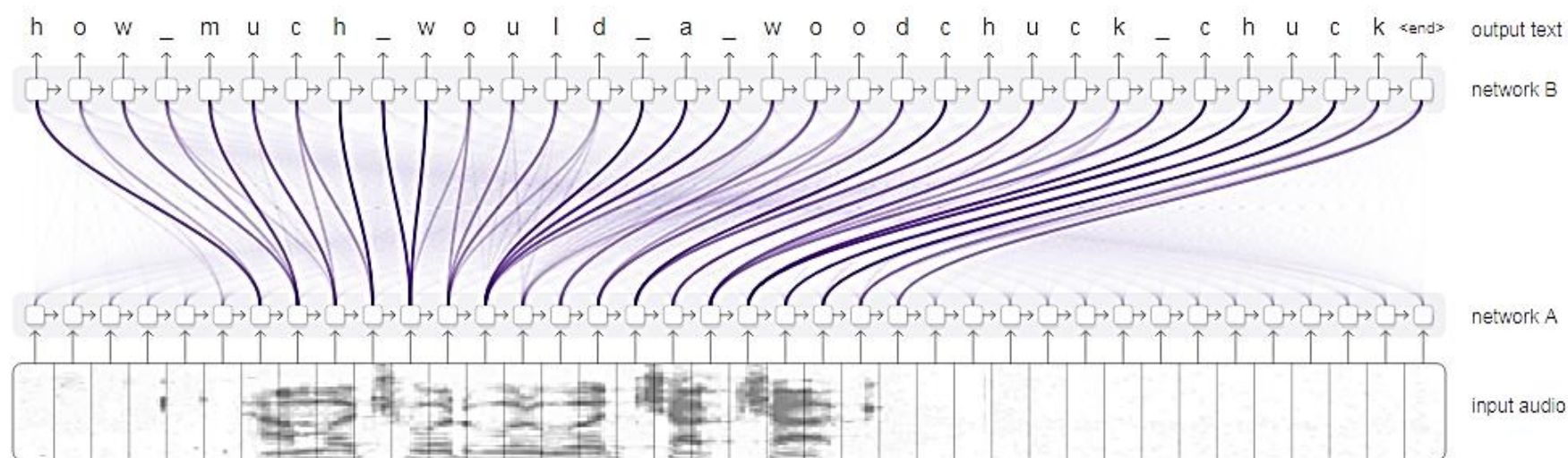
DEEP LEARNING FOR SPEECH

- The first breakthrough results of “deep learning” on large datasets happened in speech recognition
- Context-Dependent Pre-trained Deep Neural Networks for Large Vocabulary Speech Recognition, Dahl et al. (2010)

Acoustic model	Recog \ WER	RT03S FSH	Hub5 SWB
Traditional features	1-pass -adapt	27.4	23.6
Deep Learning	1-pass -adapt	18.5 (-33%)	16.1 (-32%)

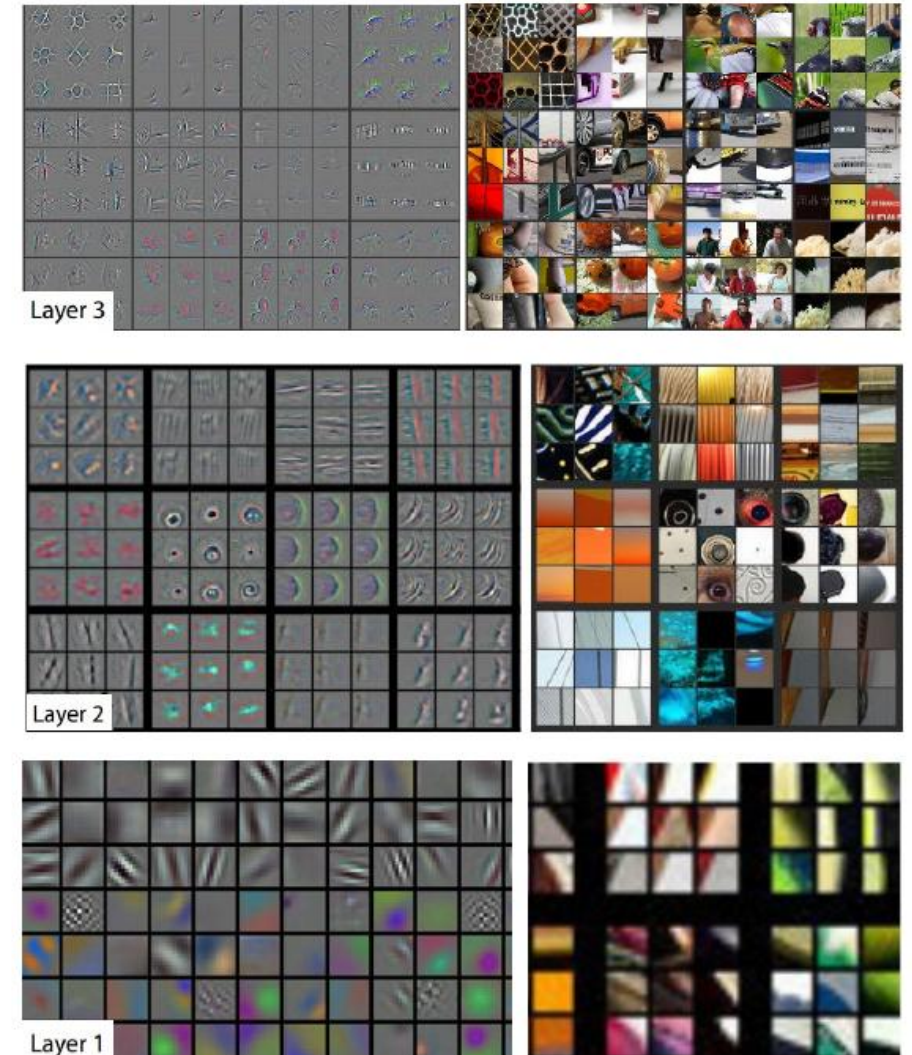


DEEP LEARNING IN ASR



DEEP LEARNING FOR COMPUTER VISION

- Most deep learning groups have (until 2 years ago) focused on computer vision
- Break through paper: ImageNet Classification with Deep Convolutional Neural Networks by Krizhevsky et al. 2012



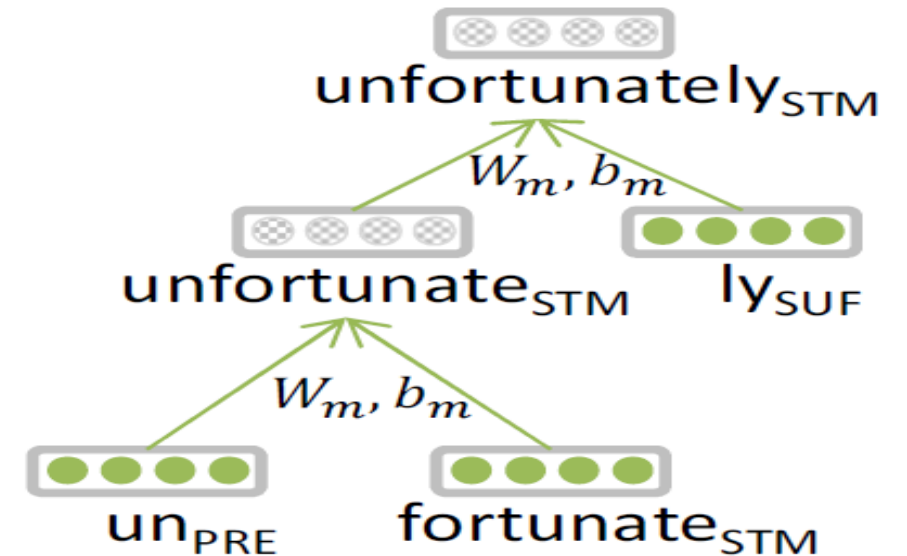
Zeiler and Fergus (2013)

REPRESENTATIONS AT NLP LEVELS: MORPHOLOGY

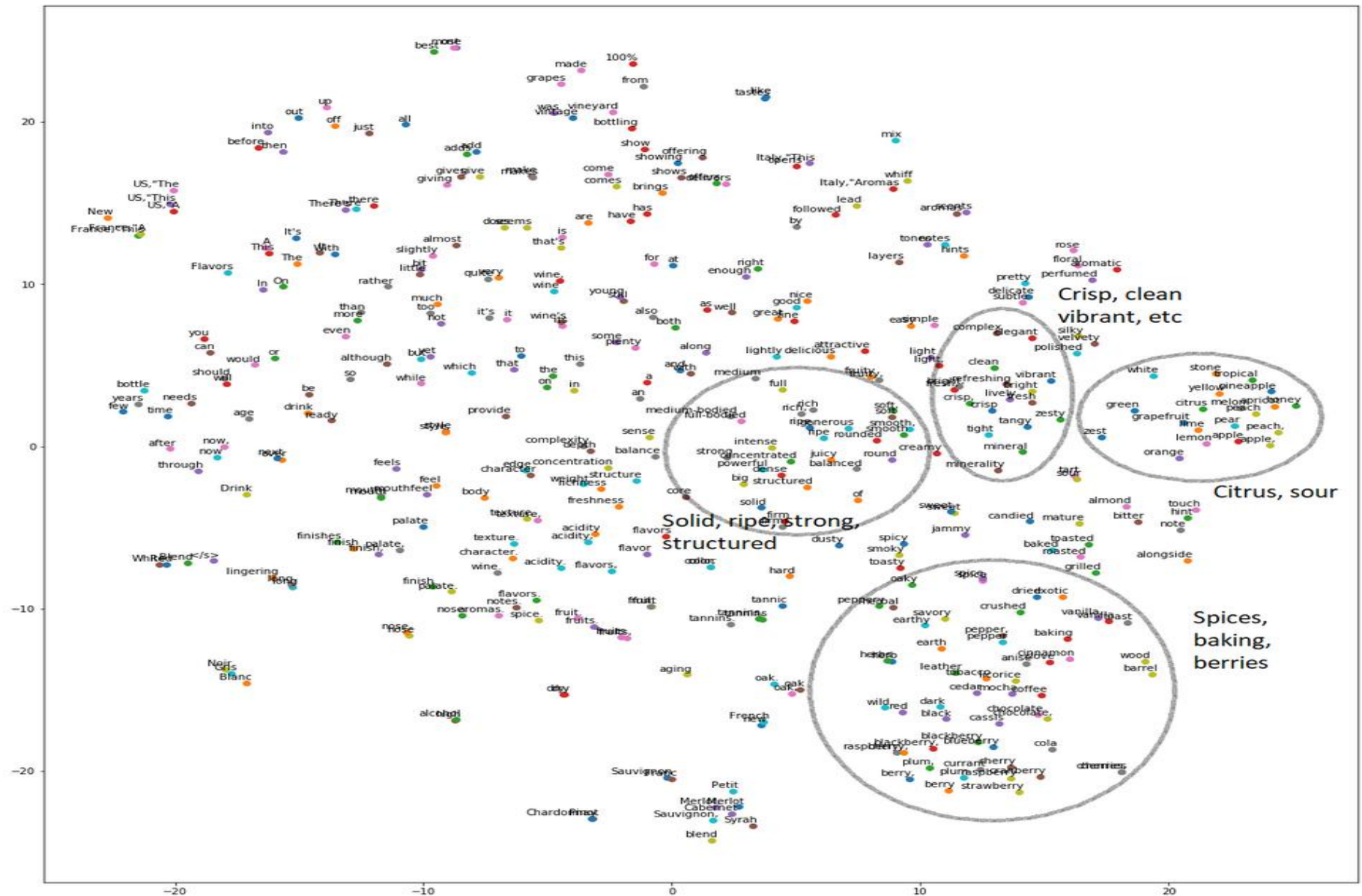
- Traditional: Morphemes

prefix	stem	suffix
un	interest	ed

- DL:
 - every morpheme is a vector
 - a neural network combines two vectors into one vector
 - Thang et al. 2013

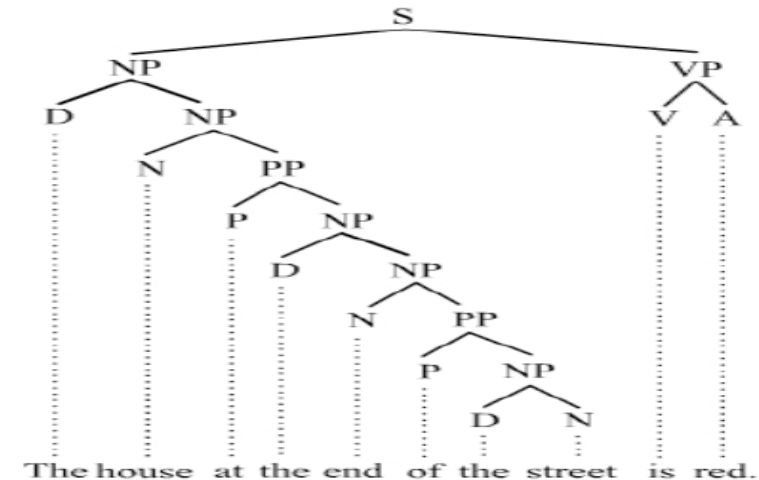


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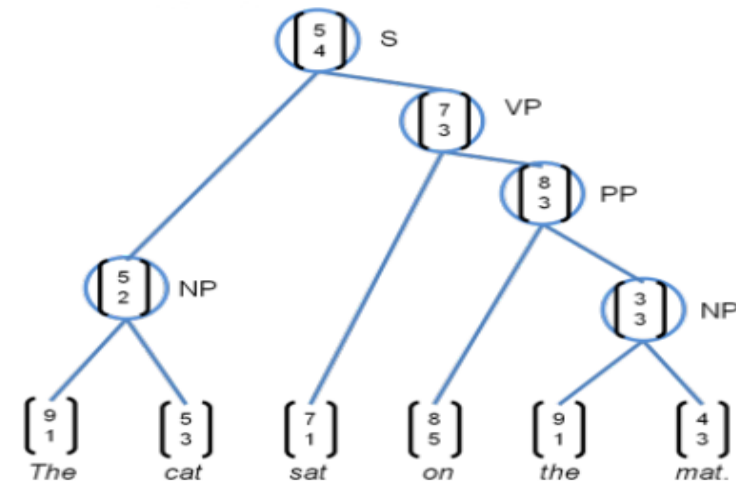


REPRESENTATIONS AT NLP LEVELS: SYNTAX

- Traditional: Phrases
Discrete categories like NP, VP

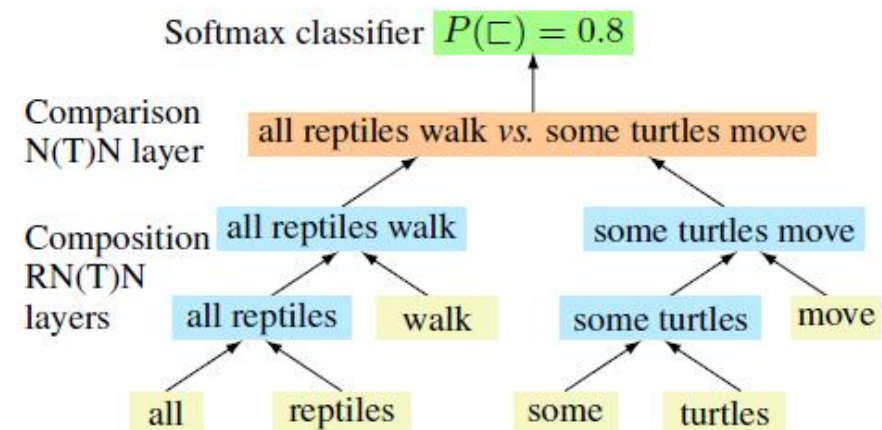
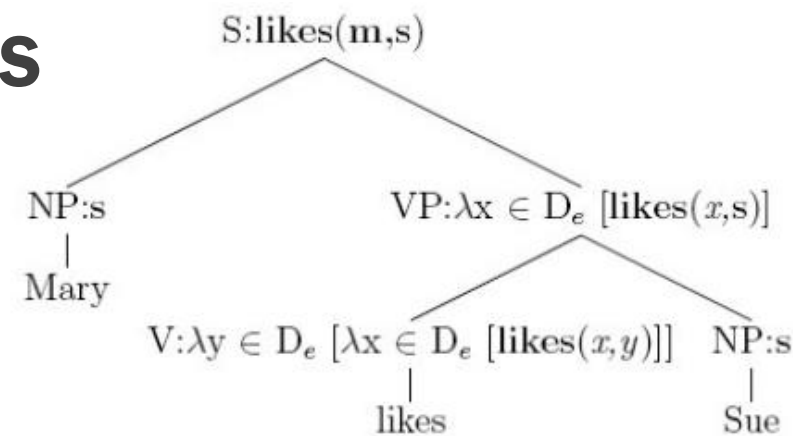


- DL:
 - Every word and every phrase is a vector
 - a neural network combines two vectors into one vector
 - Socher et al. 2011



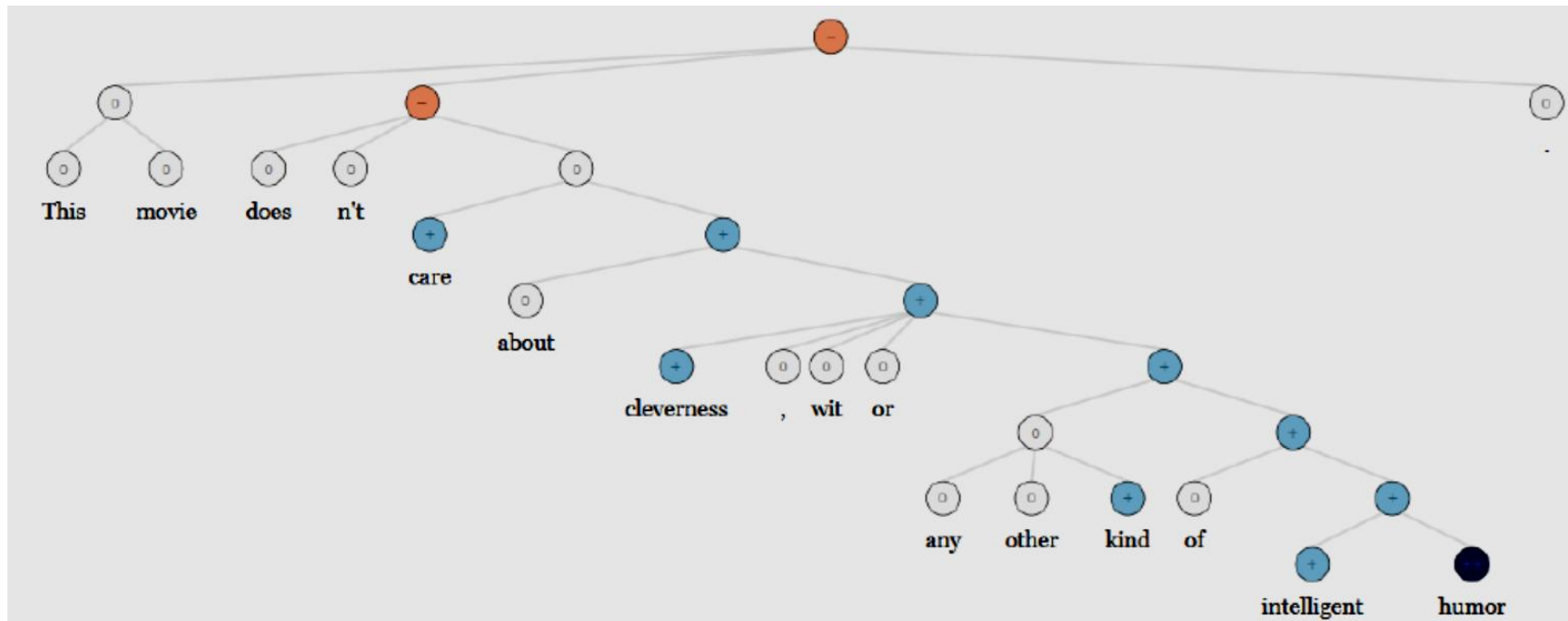
REPRESENTATIONS AT NLP LEVELS: SEMANTICS

- Traditional: Lambda calculus
 - Carefully engineered functions
 - Take as inputs specific other functions
 - No notion of similarity or fuzziness of language
- DL:
 - Every word and every phrase and every logical expression is a vector
 - a neural network combines two vectors into one vector
 - Bowman et al. 2014



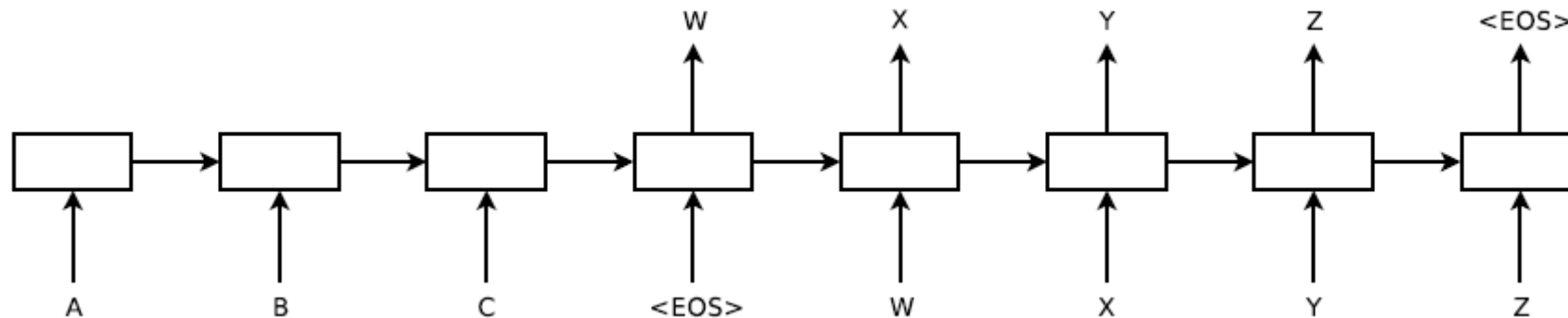
NLP APPLICATIONS: SENTIMENT ANALYSIS

- Traditional: Curated sentiment dictionaries combined with either bag-of-words representations (ignoring word order) or hand-designed negation features (ain't gonna capture everything)
- Same deep learning model that was used for morphology, syntax and logical semantics can be used! → RecursiveNN



MACHINE TRANSLATION

- Source sentence mapped to vector, then output sentence generated.



- Sequence to Sequence Learning with Neural Networks by Sutskever et al. 2014; Luong et al. 2006
- About to replace very complex hand engineered architectures.

THANK YOU

– SEE YOU TOMORROW

Dr. Pinaki Bhaskar

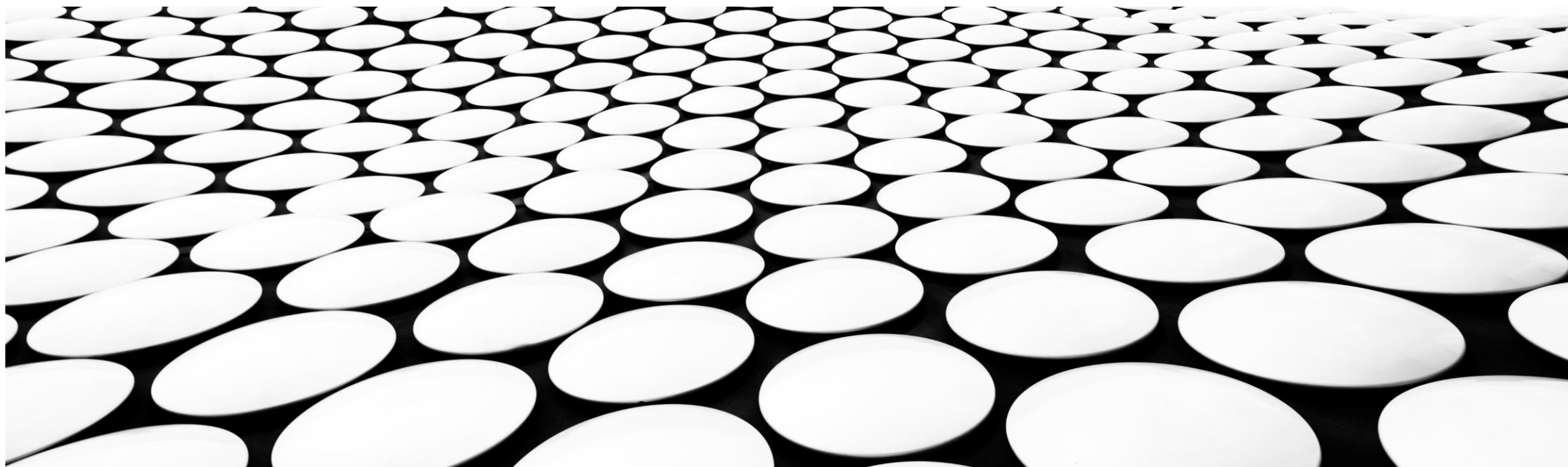
Samsung R&D Institute India,
Bangalore

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WHEN MACHINES LEARNED TO FIND MEANING ACROSS LANGUAGES

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MACHINE TRANSLATION IN BENGALI: HISTORY, METRICS, AND THE AI FRONTIER – DAY 4





THE NEURAL TURN: HOW TODAY'S SYSTEMS TRANSLATE

- From handcrafted rules to data-driven neural models
- Bengali MT now powered by large neural architectures
- Today: conceptually unpack these systems, then critique their outputs

FOUR BROAD ERAS OF MACHINE TRANSLATION: A QUICK TOUR

- Rule-based MT: linguistic rules, dictionaries, grammar transfer.
- Statistical MT: translation as probability over aligned bilingual corpora.
- Machine-learning hybrids: feature engineering, reranking, and structured models between classic SMT and neural systems.
- Deep learning / NMT: sequence-to-sequence, attention, transformers.

Short Bengali rule-based flavour example:

- Rule: “X করিতেছে” ↔ “is doing X”



WHAT IS RULE-BASED MT

- Rule-based MT begins with the assumption that language structure can be hand-described through grammar rules, bilingual lexicons, and transfer rules.
- It works by analysing the source sentence, transforming it into an intermediate or target-side structure, and generating output through handcrafted rules.
- It is especially useful when data is scarce but linguistic expertise is available.

Example:

RULE-BASED MT TYPES: DIRECT, TRANSFER, INTERLINGUA

- **Direct MT:** word-to-word substitution with minimal grammar adjustment.
- **Transfer-based MT:** source parse -> structural transfer -> target generation.
- **Interlingua MT:** source converted into a language-independent semantic representation before generation.

Example:

- Bengali: “সে আজ স্কুলে যায়নি।”
- Direct output: “He today school went-not.”
- Better transfer output: “He did not go to school today.”
- This shows why morphology and syntax matter beyond bilingual dictionaries.



RULE-BASED MT FOR BENGALI: STRENGTHS AND LIMITS

- Strengths: transparent rules, interpretable errors, useful in narrow domains, can preserve honorific distinctions if explicitly encoded.
- Limits: hard to maintain, brittle with ambiguity, expensive to scale, weak on idioms and free variation.
- Bengali challenges include rich inflection, word-order flexibility, dropped subjects in context, and variation between formal and colloquial registers.



EXAMPLE-BASED MT AND TRANSLATION MEMORY

- Between pure rule systems and data-driven systems, example-based MT used stored bilingual sentence fragments and analogical matching.
- This is close to how human translators use translation memory: “have I seen something like this before?”
- For Bengali legal or administrative text, example-based systems often worked better than naive rule systems when formulaic expressions repeated.

WORKED RULE-BASED BENGALI-ENGLISH EXAMPLE

- Source: “আজ বিকেলে হালকা বৃষ্টি হওয়ার সম্ভাবনা আছে।”
- Lexical mapping: আজ=today, বিকেলে=in the afternoon/evening, হালকা=light, বৃষ্টি=rain, সম্ভাবনা=possibility, আছে=there is.
- Rule-based output 1: “Today in afternoon light rain possibility is.”
- Rule-based output 2 with transfer rules: “There is a possibility of light rain this afternoon.”
- Fluency depends on structural transfer, not just lexical lookup.



STATISTICAL MT, HYBRID LEARNING, AND EVALUATION

- How translation became a problem of probabilities
- Why aligned corpora changed everything
- Why Bengali-English and Bengali-Hindi behave differently



FROM RULES TO PROBABILITIES

- SMT treats translation as choosing the most probable target sentence given a source sentence.
- Instead of writing grammar rules by hand, the system learns translation correspondences from parallel corpora.
- This made MT more scalable when bilingual data became available.



CORE COMPONENTS OF PHRASE-BASED SMT

- Parallel corpus
- Word alignment, often via GIZA++
- Phrase table with probabilities
- Target language model, often via KenLM
- Decoder that searches for the best translation hypothesis

SMT PIPELINE EXAMPLE

- Source: “সে আজ স্কুলে যায়নি।”
- Aligned phrases might learn: “সে”->“he/she”, “আজ”->“today”, “স্কুলে”->“to school”, “যায়নি”->“did not go”.
- Language model then prefers “He did not go to school today” over broken word order.
- This is the key advantage of SMT over naive rule substitution.



WHY SMT WAS A BREAKTHROUGH

- Required less manual grammar engineering than RBMT.
- Benefited directly from more bilingual data.
- Produced better general-purpose output for many language pairs before NMT took over.
- For Indian languages, transliteration modules and preposition handling were often added to improve quality.

EXAMPLES OF PARALLEL CORPORA

- He didn't dare the lions to attack him.
- He died in July .
- He drinks a litre of wine daily.
- He dropped off.
- He dropped out of school.
- He drove at sixty kilometers an hour.
- He earned this right as a reward for battle distinctions in Peshawar.
- He earns lots of money.
- He earns thrice as many rupees as you do.
- He eats like a pig.
- He eluded them repeatedly and while fleeing took refuge at Kedarnath in the form of a bull.
- He embarked on the construction of the massive Agra Fort in 1.
- He entered the church
- He erected a small mud fort and constructed Gavi Gangadhareswara Temple at Gavipuram and Basava Temple at Basavangudi.
- He failed in his attempt
- He fell asleep
- He fell asleep.
- He fell ill.
- He fell to thinking.
- He first built the Mehrangarh Fort and then fortified the city surrounding it.
- He fought with the Mughals on one side and on the other sultans of Bijapur.
- He founded the city in 11.
- He gave a wrong answer
- He gave her cat food. (Is he giving cat food to her or is he giving her cat some food?)

- সে সিংহকে আক্রমণ করতে ভয় পায় না।
- সে জুলাই মাসে মারা গেল।
- সে প্রত্যহ এক লিটার ওয়াইন পান করে।
- সে সরে পরেছিল।
- সে স্কুল থেকে বহিষ্কৃত হয়েছিল।
- সে মোটর গাড়ি প্রতি ঘন্টায় ৬০কিলমিটার বেগে চালিয়েছিল।
- সে এই অধিকার পুরস্কার হিসেবে অর্জন করেছিল পেশবারের যুদ্ধ পৃথকীকরণের জন্য ।
- সে অনেক অর্থ উপার্জন করে।
- সে ভোমার থেকে তিন গুণ টাকা অধিক উপার্জন করে।
- সে স্করের মত করে খাবার খাই।
- সে বার বার এড়িয়ে চলত এবং পালাতে পালাতে ষাড়ের বেশে কদারনাথে আগ্রয় নিয়েছিল।
- ১-এ সে লতুন করে বিশাল আগ্রা ফোর্টের নির্মাণ কাজ শুরু করেছিল।
- সে চার্চে প্রবেশ করল।
- সে একটা ছোটো মাটির দুর্গ বানিয়েছিল এবং গান্ধী গান্ধীপুরমে গঙ্গাধরেশ্বর মন্দির ও বসন্তগুরিতে বাসব মন্দির প্রতিষ্ঠা করেছিল।
- সে তার প্রচেষ্টায় বিফল হয়েছিল।
- সে ঘুমিয়ে পড়েছিল।
- সে ঘুমিয়ে পড়েছিল।
- সে অসুস্থ হয়ে পরেছিল।
- সে ভাবতে লেগেছিল।
- সে প্রথমে মেহরাপগড় দুর্গ বানিয়েছিল এবং তারপর তার চারপাশের বেষ্টিত শহরকে সুরক্ষিত করেছিল।
- সে একদিকে মোঘল এবং অন্যদিকে বিজাপুরের সুলতানদের সাথে লড়াই করেছিল।
- সে ১১-ই শহরটাকে খুঁজে পেয়েছিল।
- সে একটা ভুল উত্তর দিয়েছিল।
- সে তাকে বেড়ালের খাবার দিয়েছিল। (সেকি তাকে বেড়ালের খাবার দিচ্ছিল নাকি সে তার বেড়ালকে কিছু খাবার দিচ্ছিল?)



SMT PROBLEMS FOR BENGALI-ENGLISH

- Rich morphology and sparse data reduce phrase coverage.
- OOV words hurt translation quality, especially names, borrowed words, and inflected forms.
- Bengali-English has script difference plus strong syntactic divergence, which increases reordering difficulty.



BENGALI-HINDI VS BENGALI-ENGLISH

- Bengali and Hindi are both Indo-Aryan and share many cognates, though they use different scripts.
- This often makes lexical and structural transfer easier than Bengali-English in certain domains.
- However, false friends, morphological mismatches, and honorific/register issues still remain important.



SIMPLE SENTENCE COMPARISON FROM JADAVPUR WORK

- A 2018 comparison over English-Hindi and English-Bengali simple sentences found that NMT could outperform SMT on simple sentences, while SMT remained stronger over mixed-complexity corpora in that setup.
- The paper trained Moses SMT plus character-level and word-level NMT models and evaluated with BLEU, TER, adequacy, and fluency.
- This is a useful teaching example because it shows that “newer” does not automatically mean “better” under low-resource constraints.

MACHINE LEARNING BETWEEN SMT AND NMT: WHAT DO WE MEAN BY “MACHINE-LEARNING-BASED MT”?

- Feature-engineered reranking and discriminative learning on top of SMT.
- Structured prediction for reordering and phrase selection.
- Neural language models inserted into SMT pipelines before full end-to-end NMT became dominant.
- These hybrids matter historically because they are the bridge between classical SMT and deep learning.



HYBRID BENGALI-ENGLISH WORK

- A 2019 BUET thesis explicitly studied blending rule-based translation with SMT and NMT for Bengali-to-English translation.
- The goal was to use rule knowledge for grammar-sensitive phenomena while leveraging data-driven systems for fluency and coverage.
- This is pedagogically useful because it shows that hybrid systems often emerge when no single paradigm fully solves the language pair.



WHY HYBRID SYSTEMS MAKE SENSE FOR BENGALI

- Rules can encode honorifics, verb agreement patterns, and named-entity handling.
- SMT contributes phrase probabilities and fluent target-side preferences.
- Neural systems contribute context-sensitive generation.
- Bengali often benefits from such layering because of limited domain-balanced resources.



WHY DEEP LEARNING CHANGED MT

- Early neural ideas existed before the transformer era, but they were constrained by data and compute.
- Neural models first improved components inside SMT, then replaced the whole pipeline.
- By 2016–2017, neural MT had overtaken SMT in most major evaluations.



SHALLOW MACHINE LEARNING VS DEEP LEARNING

- Classic machine learning usually relies on engineered features.
- Deep learning learns internal representations automatically from data.
- For MT, this meant moving from phrase tables and manual modules to end-to-end sentence modeling.



FEEDFORWARD INTUITION

- A feedforward network maps input vectors to output vectors through stacked hidden layers.
- By itself, it is not ideal for sequence generation because it lacks recurrence or explicit sequence modeling.
- Still, feedforward networks appear inside larger MT systems as scoring components and projection layers.



WHY RNNS MATTERED

- RNNs introduced recurrence, letting the model process sequences with memory over previous tokens.
- This was a major step for translation because language is sequential.
- But vanilla RNNs struggled with long dependencies and vanishing gradients.



LSTM AND GRU

- LSTMs use gated memory cells to preserve relevant long-range information.
- GRUs are a lighter gated alternative with fewer parameters.
- Both became popular in early NMT for Indian languages, including Bengali-English and Bengali-Hindi.



SEQ2SEQ MODEL

- Encoder reads the source sentence.
- Decoder generates the target sentence token by token.
- This was revolutionary because the entire translation process became one trainable



PROBLEM OF FIXED VECTORS

- Early seq2seq forced the entire source sentence into one fixed-length vector.
- This worked poorly for long or complex sentences.
- Attention solved this bottleneck.



ATTENTION MECHANISM

- Attention lets the decoder focus on different parts of the source sentence at each output step.
- This improved long-sentence handling and contextual alignment.
- It also made model behavior somewhat more interpretable via alignment heatmaps.



BENGALI-HINDI ATTENTION-BASED NMT

- Das et al. 2016 studied attention-based NMT for the Hindi-Bengali pair and analysed error types in a low-resource, morphologically rich setting.
- They also explored heuristic post-processing to improve outputs.
- This paper is excellent for explaining why Indic low-resource conditions shaped neural MT differently than high-resource European pairs.

EXAMPLE: BENGALI TO HINDI WITH ATTENTION

Source Bengali:

- “সে ধীরে ধীরে ঘরের দিকে হাঁটল।”

Possible Hindi:

- “वह धीरे-धीरे घर की ओर चला।”

Attention helps align “ধীরে ধীরে” with “धीरे-धीरे” and “ঘরের দিকে” with “घर की ओर”, rather than compressing everything into one fixed vector.



BYTE PAIR ENCODING AND SUBWORDS

- Subword segmentation reduces the rare-word problem by splitting words into frequent pieces.
- This is especially useful for morphologically rich and low-resource languages.
- It improves coverage of Bengali inflected forms and borrowed words.



TRANSFORMER REVOLUTION

- The transformer replaced recurrence with self-attention.
- It allowed more parallel training and stronger long-range modeling.
- This architecture now dominates MT and underlies large language models too.

TRANSFORMERS: READING AND WRITING WITH ATTENTION

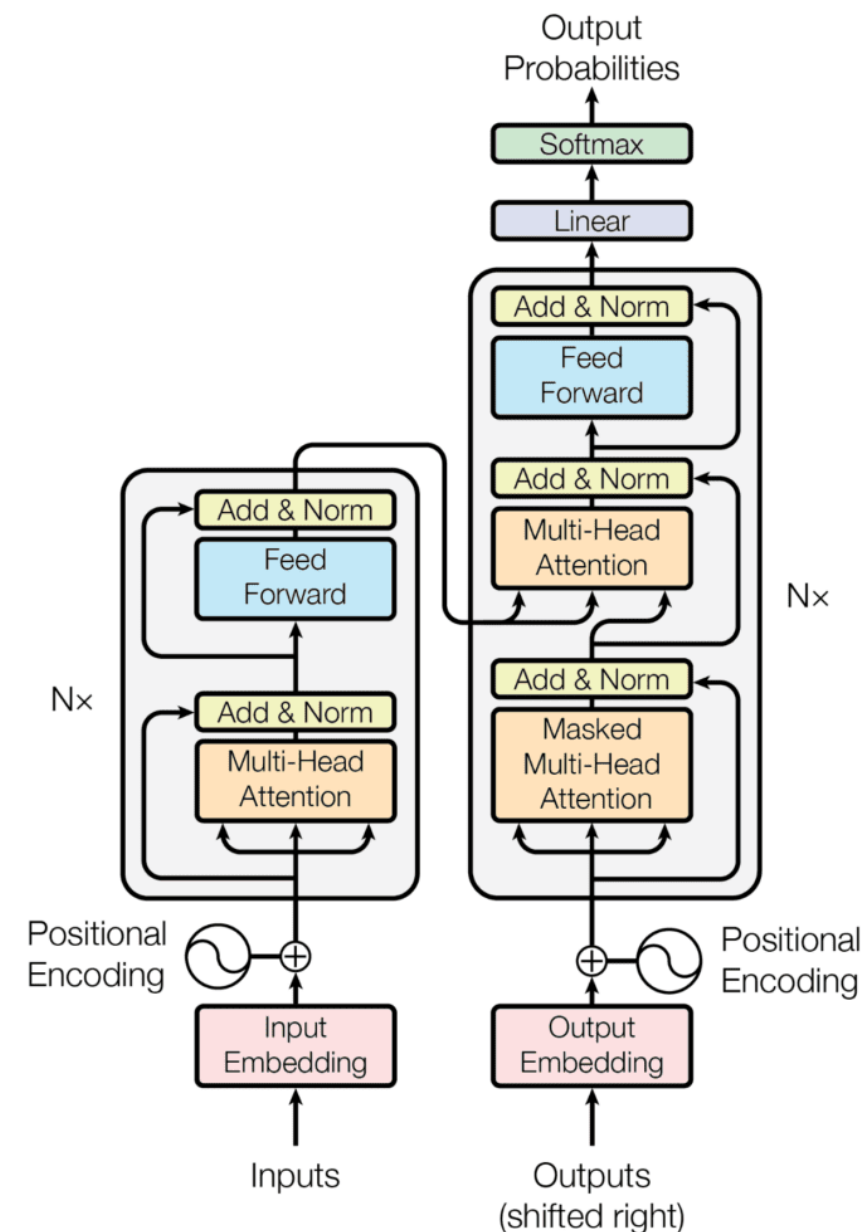
- Encoder: reads entire source sentence and builds contextual representations
- Decoder: writes target sentence one token at a time, using this context
- Attention: at each step, model decides **which source words to focus on**
- Think of a human translator moving eyes back and forth over the source paragraph

Example source snippet:

- “আজ বিকেলে হালকা বৃষ্টি হওয়ার সম্ভাবনা আছে।”
Rough output: “There is a chance of light rain this evening.”

DETAILS OF TRANSFORMER ARCHITECTURE

- **Encoder and Decoder Structure:** The transformer generally consists of an encoder (to read/process input) and a decoder (to generate output), both composed of stacked layers.
- **Self-Attention Mechanism:** This allows the model to look at other words in the input sentence to gain a better understanding of the current word, allowing it to capture context efficiently.
- **Multi-Head Attention:** Instead of a single attention pass, the model runs multiple self-attention mechanisms ("heads") in parallel to focus on different positions and aspects of the information.
- **Positional Encoding:** Because transformers process tokens in parallel, they do not inherently understand word order. Positional encoding adds information about the position of each word in the sequence.
- **Feed-Forward Networks:** Each encoder/decoder layer contains a feed-forward network to process the attention outputs independently.
- **Residual Connections & Normalization:** Residual connections around each sub-layer (attention and feed-forward) are followed by layer normalization to aid in training stability and deep model optimization.





WHAT HAPPENS WHEN YOU CLICK “TRANSLATE”

- Input text is normalised and split into subword units
- Encoder maps subwords to vectors and builds contextual representations
- Decoder generates target tokens step by step, using attention to focus on the right parts of the source
- Output is detokenised, punctuated, and sometimes post-edited by simple rules

Example pipeline sentence:

- Bengali: “তুমি কি কাল দুপুরে সময় পাবে?”
- System output: “Will you be free tomorrow afternoon?”

INDICTRANS2: NEURAL MT FOR INDIAN LANGUAGES

- Open-source transformer MT covering 22 scheduled Indian languages, incl. Bengali
- Trained on large parallel corpora (BPCC and others) with unified script representations
- Supports English ↔ Bengali and many Indic↔Indic directions
- It uses script unification where feasible to improve transfer learning across Indic languages.
- A key resource for researchers and practitioners working with Bengali MT

Example:

- English input: “Cyclone Amphan caused severe damage in coastal Bengal.”
- IndicTrans2 output (approx.): “ঘূর্ণিঝড় আমফান উপকূলীয় বাংলায় ভয়াবহ ক্ষতি করেছে।”



WHY INDICTRANS2 MATTERS FOR BENGALI-ENGLISH AND BENGALI-HINDI

- One model can support English↔Bengali and Bengali↔Hindi within a multilingual framework.
- This is especially helpful when some pairs are low-resource, because shared learning across languages improves robustness.
- It reflects the current state of open Indian-language MT infrastructure.

BENGALI LANGUAGE MODELS: BANGLABERT & BEYOND

- BanglaBERT: transformer model trained on ~27.5 GB of Bengali text
- Strong performance on sentiment, NER, QA, and other NLP benchmarks
- Other Bengali BERT variants: Bangla-BERT, banBERT, etc.
- Mostly used for **understanding** Bengali; NMT combines such representations with generation

Example classification task:

- Sentence: “ছবিটা দারুণ লেগেছে, আবার দেখতে ইচ্ছে করছে।” → positive sentiment



BANGLA-ENGLISH NEURAL MT REFERENCES

- A 2021 paper reported attention-based Bangla↔English MT using GRU/LSTM-style sequence modeling, with BLEU values around 21.42 for Bangla→English and 25.44 for English→Bangla in that study's setting.
- A 2024 survey reviewed many Bangla-English NMT systems and grouped them into transformers, RNNs with enhancements, and attention-based RNNs.
- These studies are useful comparatively, though their datasets and settings vary greatly, so headline BLEU numbers should not be compared naively.

BENGALI-ENGLISH NEURAL EXAMPLE

- Source:
“কলকাতার রাস্তায় আজ অস্বাভাবিক ভিড় দেখা গেছে।”
- Possible NMT output:
“Unusual crowds were seen on the streets of Kolkata today.”
- This is better than literal output like “On Kolkata’s roads today abnormal crowd was seen.”
- Neural systems improve fluency by modelling context across the full sentence.

BENGALI-HINDI NEURAL EXAMPLE

- Source:
“আজ শহরের রাস্তায় অস্বাভাবিক ভিড় দেখা গেছে।”
- Possible Hindi output:
“आज शहर की सड़कों पर असामान्य भीड़ देखी गई।”
- Lexical similarity and parallel structure can make this pair easier in formal prose, while colloquial or literary text can still be difficult.



LIMITS OF NEURAL MT

- Rare words and unseen phrases remain a challenge.
- Ambiguity and culturally specific meaning remain hard.
- Training is resource-intensive and data-hungry.
- Low-resource languages like Bengali still face domain imbalance.

WHERE NEURAL MT WORKS WELL – AND WHERE IT DOESN'T

- Works well: news, factual prose, short commands, everyday sentences
- Struggles: poetry, humour, sarcasm, dialects, politeness levels
- Data biases: corpora over-represent news, Wikipedia, official texts
- Question: whose Bengali does the model speak, and whose does it ignore?

Contrast example:

- Colloquial: “ওরে, আজকে কিন্তু একদম জমে গেল রে!”
- System tends to normalise: “আজ কিন্তু অনেক ভালো ছিল।” – tone softens.



WHY COMPARISON MATTERS

- MT quality depends on language pair, domain, sentence complexity, and evaluation metric.
- No single paradigm wins uniformly across all settings.
- Humanistic critique is essential because “better BLEU” is not the same as “better translation.”

BENGALI-ENGLISH: WHAT USUALLY GOES WRONG

- Articles: “a/the” absent in Bengali but required in English.
- Tense/aspect nuance often shifts.
- Idioms become too literal.
- Formal vs colloquial Bengali may collapse into neutral English.
Use source:
- “ওঁর কথা শুনে আমার মাথা গরম হয়ে গেল।”
Literal: “My head became hot hearing his words.”
Natural: “What he said made me furious.”

BENGALI-HINDI: WHAT USUALLY GOES WRONG

- Similarity can mislead systems into over-literal mapping.
- Honorific and politeness distinctions may shift.
- Aspect and auxiliary choice can drift.
Source:
 - “আপনি একটু বসুন, আমি এখনই আসছি।”
Hindi:
 - “आप थोड़ा बैठिए, मैं अभी आता हूँ।”
Whether “आता हूँ” vs “आ रही हूँ” etc. matters under different speakers and contexts.

FORMAL PROSE EXAMPLE

- “সরকারের সাম্প্রতিক সিদ্ধান্তটি বাস্তবায়নের ক্ষেত্রে স্বচ্ছতার অভাব গভীর উদ্বেগের কারণ।”

English:

- “The lack of transparency in implementing the government’s recent decision is a matter of deep concern.”

Hindi:

- “सरकार के हालिया निर्णय के क्रियान्वयन में पारदर्शिता की कमी गहरी चिंता का कारण है।”

Adequacy, register, and compression.

LITERARY PROSE EXAMPLE

- Source:
“দিগন্ত জুড়ে লাল আভা, যেন শহরটা এক অদ্ভুত স্বপ্নের ভিতরে ঢুকে গেছে।”
- English options:
“A red glow spread across the horizon, as if the city had slipped into a strange dream.”
“The horizon was washed in red, as though the city had entered an uncanny dream.”
- Hindi option:
“क्षितिज पर लाल आभा फैली थी, मानो शहर किसी अजीब सपने में उतर गया हो।”
- Which version preserves imagery best?

CONVERSATIONAL EXAMPLE

- Source:
- A: “তুই কাল আসছিস তো?”
- B: “দেখি, অফিসে কী হয়।”
- English:
- “You’re coming tomorrow, right?” / “Let’s see what happens at the office.”
- Hindi:
- “तू कल आ रहा है न?” / “देखते हैं, ऑफिस में क्या होता है।”
Discuss intimacy, informality, and whether systems flatten “তুই” into a neutral “you”.



BREAK

FOR 10 MINS





METRICS VS HUMAN JUDGEMENT

- A 2023 comparison cited Bengali-Hindi attention-based NMT outperforming MOSES SMT on that pair in the referenced study.
- But metric gains do not automatically imply better handling of literary tone, register, or cultural nuance.
- Always pair automatic metrics with human review, especially in Bengali literary or pedagogical contexts.



DESIGNING A HUMAN-CENTRED EVALUATION RUBRIC

- Automatic scores help, but human evaluation is the gold standard
- Proposed criteria (1–5 scale each):
 - Adequacy (does it convey the meaning?)
 - Fluency (does it sound natural in target language?)
 - Register & politeness
 - Cultural nuance & metaphor
 - Bias & harmful distortions
- We will adapt and extend this rubric together



POST-EDITING DEMONSTRATION

- Machine output:
 - “Today on city roads abnormal crowd was seen.”
- Post-edited:
 - “Unusually heavy crowds were seen on city streets today.”
- It shows that human correction often improves article use, adjective choice, and natural collocation.

COMPARING PARADIGMS ON ONE SENTENCE

- Source:
 - “সে আজ স্কুলে যায়নি, কারণ তার জ্বর ছিল।”
- Rule-based:
 - “He today school not-went because his fever was.”
- SMT-like:
 - “He did not go to school today because he had fever.”
- Neural:
 - “He did not go to school today because he had a fever.”
- Hindi:
 - “वह आज स्कूल नहीं गया क्योंकि उसे बुखार था।”
This single slide is very effective pedagogically.



WHY OLDER PARADIGMS STILL MATTER

- RBMT teaches structure and explicit grammar.
- SMT teaches alignment, phrase probabilities, and data dependence.
- NMT teaches contextual representation and fluent generation.
- Understanding all three helps critique current AI rather than treating it as magic.



COMPARATIVE LAB: THREE TEXT TYPES

- We will examine outputs for:
 - A short literary passage
 - A newspaper editorial paragraph
 - A conversational dialogue with code-mixing
- For each: compare outputs from multiple systems (e.g., IndicTrans2, Google Translate, ChatGPT)
- Use the rubric to score and discuss disagreements

LITERARY PASSAGE: STYLE AND AMBIGUITY

- Source (short, made-up but literary):
 - “দিগন্ত জুড়ে লাল আভা, যেন শহরটা হঠাৎ করে এক স্বপ্নের ভিতরে ঢুকে গেছে।”
- System Output A: “Red glow across the horizon, as if the city has suddenly entered a dream.”
- System Output B: “The horizon is red, as though the city has suddenly gone into a dream.”
- Discuss: which preserves mood and rhythm better? How would **you** translate it?

EDITORIAL PARAGRAPH: TONE AND STANCE

- Source (editorial-style):
 - “সরকারের সাম্প্রতিক সিদ্ধান্তটি অনেক ইতিবাচক দিক খুলে দিলেও, বাস্তবায়নের ক্ষেত্রে স্বচ্ছতার অভাব গভীর উদ্বেগের কারণ।”
- Output A: “Although the government’s recent decision has opened many positive aspects, the lack of transparency in implementation is a matter of deep concern.”
- Output B: “The government’s latest move looks good on paper, but the opaque implementation is deeply worrying.”
- Question: which captures the stance more clearly? How does tone shift?

DIALOGUE: CODE-MIXING AND HONORIFICS

- Source dialogue:
 - A: “তুমি কাল মিটিং-এ আসছ তো?”
 - B: “দেখি রে, boss যদি আরেকটা call না দেয়।”
- Output A:
 - “You are coming to the meeting tomorrow, right?”
 - “Let me see, if the boss doesn’t give another call.”
- Output B:
 - “Are you attending the meeting tomorrow?”
 - “Let’s see, unless my boss puts me on another call.”
- Discuss: register, informality, and how to handle “রে” / “boss”



WHEN NUMBERS DISAGREE WITH HUMAN JUDGMENT

- Suppose Output A has higher BLEU / chrF than Output B
- But as a group, you may prefer Output B for tone and nuance
- This is common for Bengali, especially in literary or colloquial texts
- Lesson: treat metrics as **indicators**, not final arbiters of quality



FROM SINGLE TRANSLATIONS TO AI-ASSISTED WORKFLOWS

- LLM-based agents can chain steps:
 - Retrieve relevant sources
 - Draft translations with glossary constraints
 - Check consistency and style across a corpus
- For Bengali, such workflows can support literary projects, archives, and subtitling
- Human translators remain responsible for final decisions and ethics

THANK YOU

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